

Gene Regulation

Michael Schatz

October 22 2025

Lecture 17. Applied Comparative Genomics



Project Prelim Report

Due: Monday November 3

Preliminary Project Report

Assignment Date: Wednesday October 22, 2025

Due Date: Monday, November 3, 2025 @ 11:59pm

Each team should submit a PDF of your preliminary project proposal (2 to 3 pages) to [GradeScope](#) by 11:59pm on Monday November 3

The preliminary report should have at least:

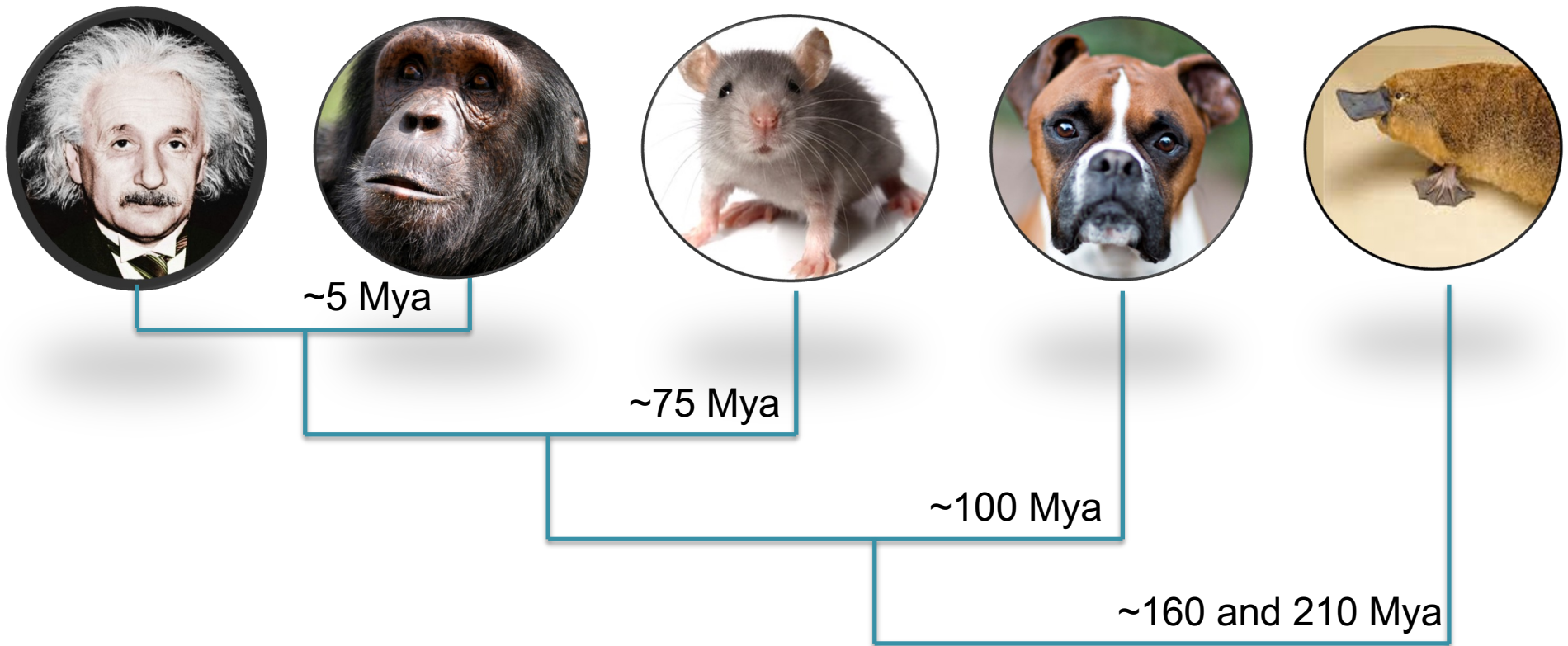
- Title of your project
- List of team members and email addresses
- 1 paragraph abstract summarizing the project
- 1+ paragraph of Introduction
- 1+ paragraph of Methods that you are using
- 1+ paragraph of Results, describing the data evaluated and any preliminary results
- 1+ paragraph of Discussion (what you have seen or expect to see)
- 1+ figure showing a preliminary result (typically a summary of the data you have identified for your project)
- 5+ References to relevant papers and data

The preliminary report must use the Bioinformatics style template. Word and LaTeX templates are available at https://academic.oup.com/bioinformatics/pages/submission_online. [Overleaf](#) is recommended for LaTeX submissions. [Google Docs](#) is recommended for non-latex submissions, especially group projects. [Paperpile](#) is recommended for citation management.

Later, you will present your project in class starting the week of November 19. You will also submit your final written report (6-8 pages) of your project by Dec 16 with a draft due on Dec 10 and a final project presentation on Dec 11-15

Please use Piazza if you have any questions!

Human Evolution

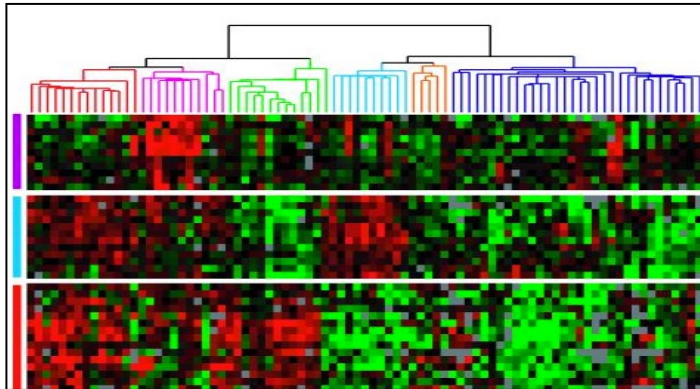


As expected, the majority of platypus genes (82%; 15,312 out of 18,596) have orthologues in these five other amniotes (Supplementary Table 5). The remaining 'orphan' genes are expected to primarily reflect rapidly evolving genes, for which no other homologues are discernible, erroneous predictions, and true lineage-specific genes that have been lost in each of the other five species under consideration.

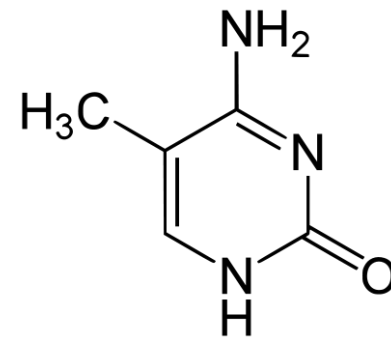
Genome analysis of the platypus reveals unique signatures of evolution
(2008) *Nature*. 453, 175-183 doi:10.1038/nature06936

*-seq in 4 short vignettes

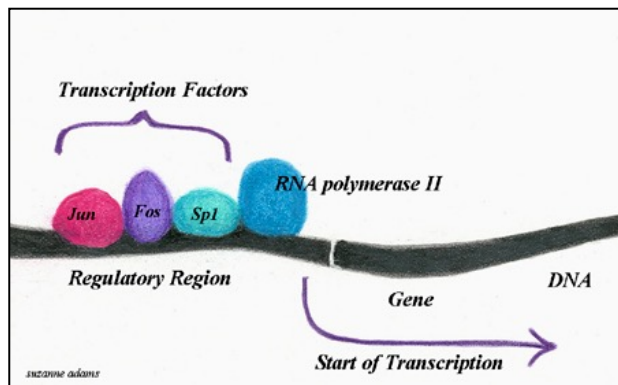
RNA-seq



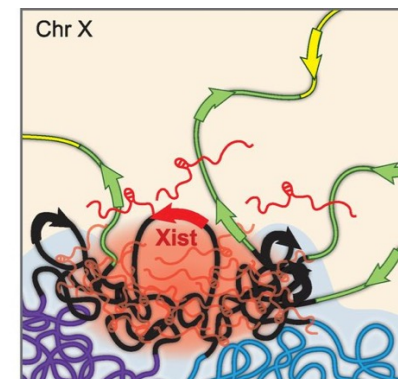
Methyl-seq



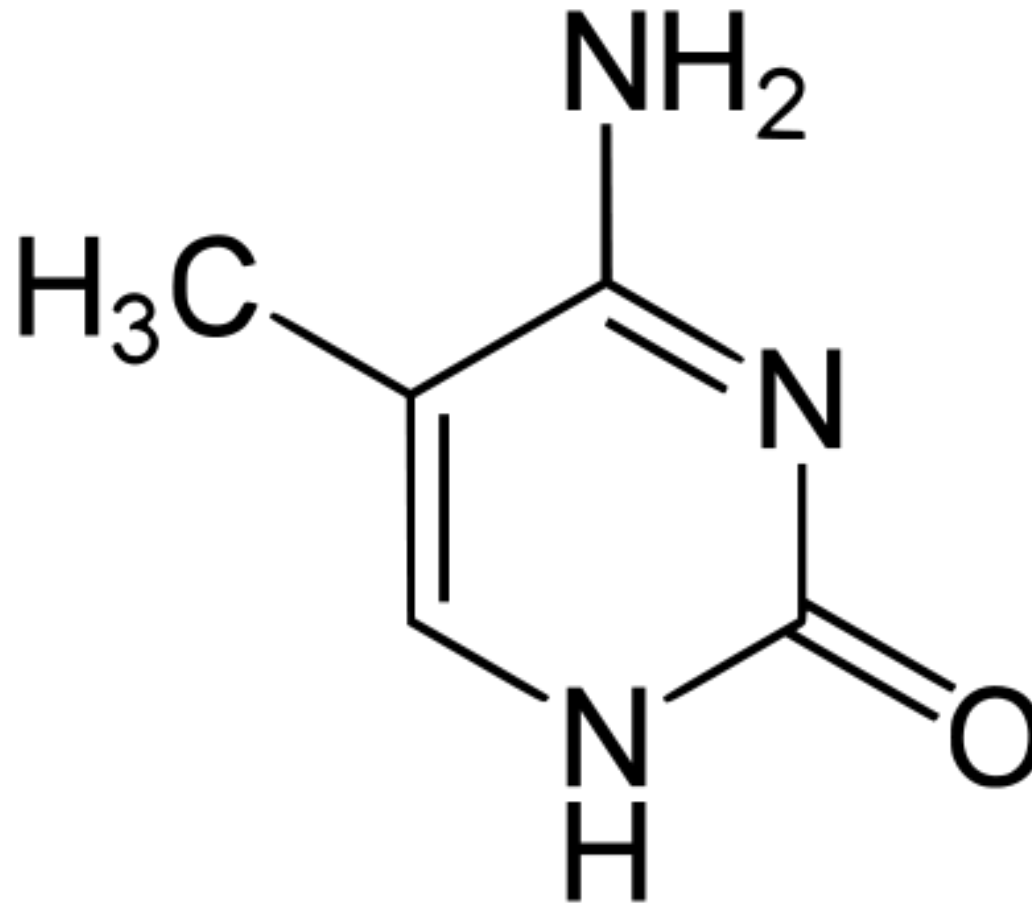
ChIP-seq



Hi-C



Methyl-seq

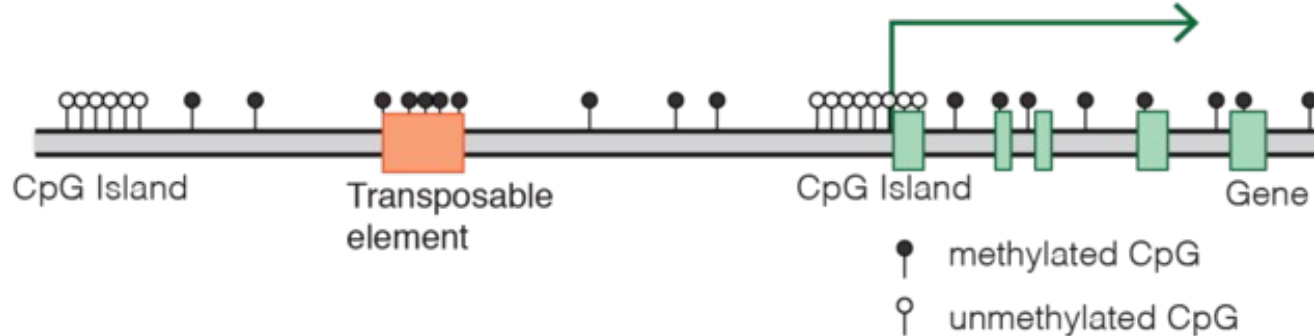


Finding the fifth base: Genome-wide sequencing of cytosine methylation

Lister and Ecker (2009) *Genome Research*. 19: 959-966

Methylation of CpG Islands

Typical mammalian DNA methylation landscape



CpG islands are (usually) defined as regions with

- 1) a length greater than 200bp,
- 2) a G+C content greater than 50%,
- 3) a ratio of observed to expected CpG greater than 0.6

Methylation in promoter regions correlates negatively with gene expression.

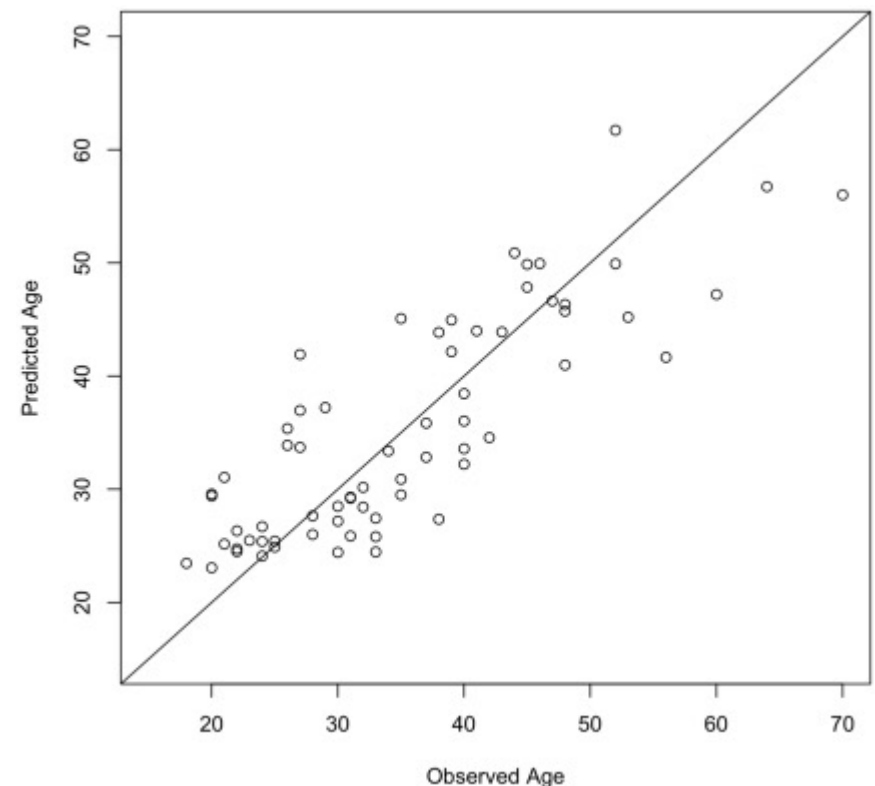
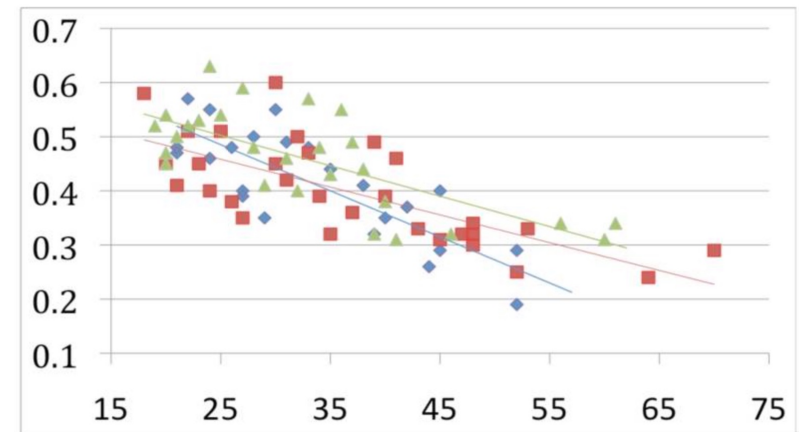
- CpG-dense promoters of actively transcribed genes are never methylated
- In mouse and human, around 60-70% of genes have a CpG island in their promoter region and most of these CpG islands remain unmethylated independently of the transcriptional activity of the gene
- Methylation of DNA itself may physically impede the binding of transcriptional proteins to the gene
- Methylated DNA may be bound by proteins known as methyl-CpG-binding domain proteins (MBDs) that can modify histones, thereby forming compact, inactive chromatin, termed heterochromatin.

Bisulfite Conversion



Bismark: a flexible aligner and methylation caller for Bisulfite-Seq applications
Krueger and Andrews (2010) *Bioinformatics*. 27 (11): 1571-1572.

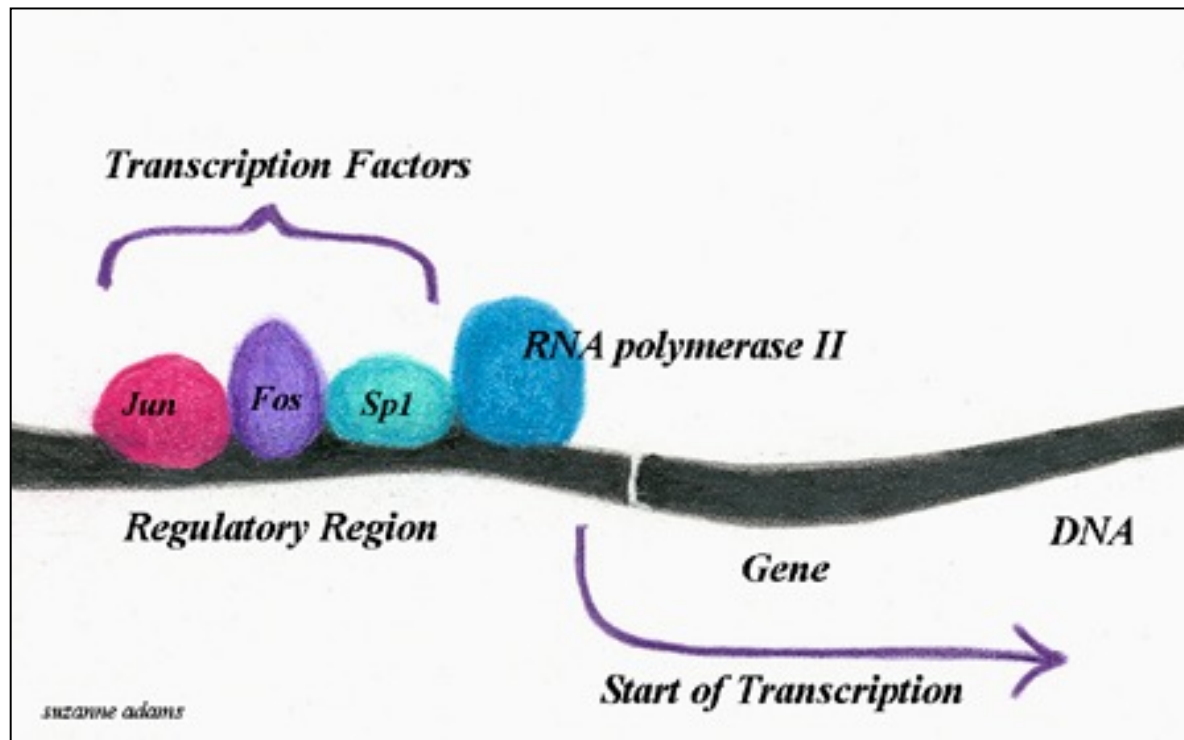
From the moment of conception, we begin to age. A decay of cellular structures, gene regulation, and DNA sequence ages cells and organisms. DNA methylation patterns change with increasing age and contribute to age related disease. **Here we identify 88 sites in or near 80 genes for which the degree of cytosine methylation is significantly correlated with age in saliva of 34 male identical twin pairs between 21 and 55 years of age.** Furthermore, we validated sites in the promoters of three genes and replicated our results in a general population sample of 31 males and 29 females between 18 and 70 years of age. The methylation of three sites—in the promoters of the EDARADD, TOM1L1, and NPTX2 genes—is linear with age over a range of five decades. **Using just two cytosines from these loci, we built a regression model that explained 73% of the variance in age, and is able to predict the age of an individual with an average accuracy of 5.2 years.** In forensic science, such a model could estimate the age of a person, based on a biological sample alone. Furthermore, a measurement of relevant sites in the genome could be a tool in routine medical screening to **predict the risk of age-related diseases and to tailor interventions based on the epigenetic bio-age instead of the chronological age.**



Epigenetic Predictor of Age

Bocklandt et al (2011) PLOS One. doi: 10.1371/journal.pone.0014821

ChIP-seq



Genome-wide mapping of in vivo protein-DNA interactions.

Johnson et al (2007) *Science*. 316(5830):1497-502

Transcription

Transcription

2,018,430 views

ndsuvirtualcell
Uploaded on Jan 30, 2008

NDSU Virtual Cell Animations Project animation 'Transcription'. For more information please see <http://vcell.ndsu.edu/animations>

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Transcription and Translation: From DNA to Protein
Professor Dave Explains
151K views
6:27

DNA - transcription and translation
Wisam Kabaha
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Transcription and mRNA processing | Biomolecules |
Khan Academy ✓
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DNA transcription and translation Animation
Haider abd
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Translation
ndsuvirtualcell
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Transcription and Translation Overview
Armando Hasudungan
611K views
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DNA, Hot Pockets, & The Longest Word Ever: Crash
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Transcription 1
khanacademymedicine ✓
263K views
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TRANSCRIPTION
congthanng
795K views
1:28

Moana - Best Scenes (FHD)

<https://www.youtube.com/watch?v=WsofH466lqk>

Transcription Factors

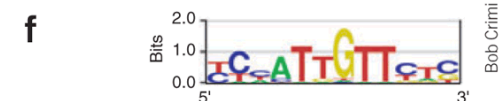
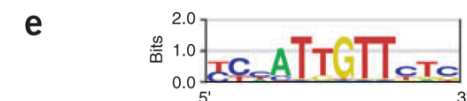
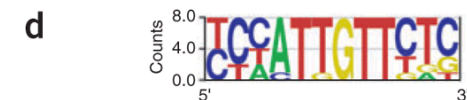
A transcription factor (or sequence-specific DNA-binding factor) is a protein that controls the rate of transcription of genetic information from DNA to messenger RNA, by binding to a specific DNA sequence.

- Transcription factors work alone or with other proteins in a complex, by promoting (as an activator), or blocking (as a repressor) the recruitment of RNA polymerase to specific genes.
- A defining feature of transcription factors is that they contain at least one DNA-binding domain (DBD)
- Figure (a) Eight known genomic binding sites in three *S. cerevisiae* genes. (b) Degenerate consensus sequence. (c,d) Frequencies of nucleotides at each position. (e) Sequence logo (f) Energy normalized logo using relative entropy to adjust for low GC content in *S. cerevisiae*.

a HEM13 CCCATTGTTCTC
 HEM13 TTTCTGGTTCTC
 HEM13 TCAATTGTTTAG
 ANB1 CTCATTGTTGTC
 ANB1 TCCATTGTTCTC
 ANB1 CCTATTGTTCTC
 ANB1 TCCATTGTTTCGT
 ROX1 CCAATTGTTTGTG

b YCHATTGTTCTC

c A 002700000010
 C 464100000505
 G 000001800112
 T 422087088261



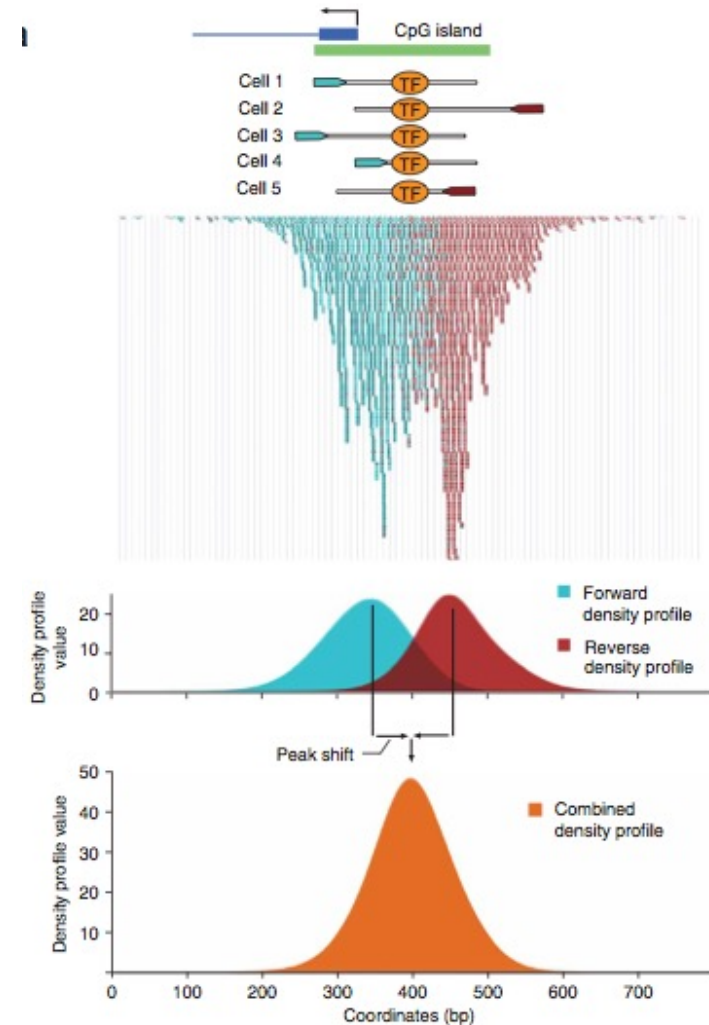
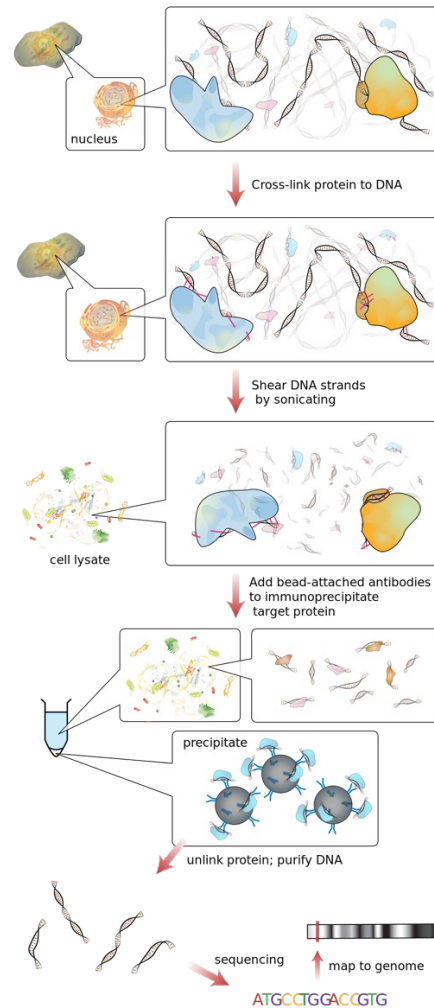
What are DNA sequence motifs?

D'haeseleer (2006) Nature Biotechnology 24, 423 – 425 doi:10.1038/nbt0406-423

ChIP-seq: TF Binding

Goals:

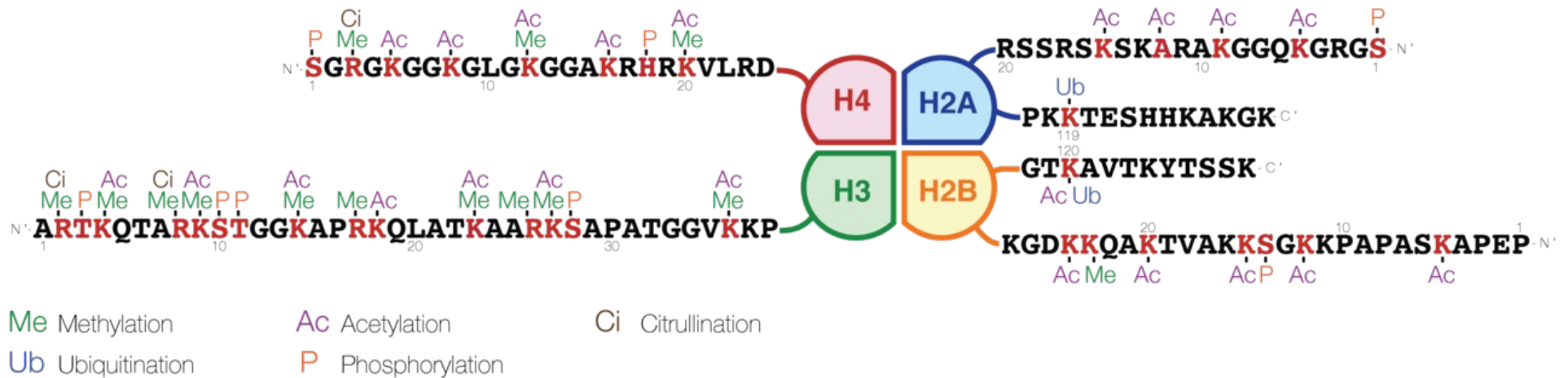
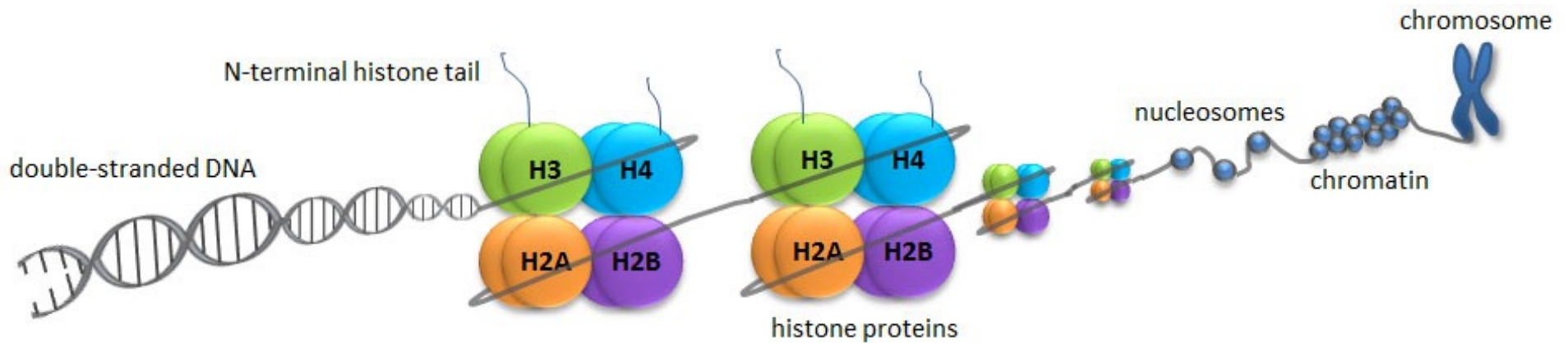
- Where are transcription factors and other proteins binding to the DNA?
- How strongly are they binding?
- Do the protein binding patterns change over developmental stages or when the cells are stressed?



Genome-wide analysis of transcription factor binding sites based on ChIP-Seq data

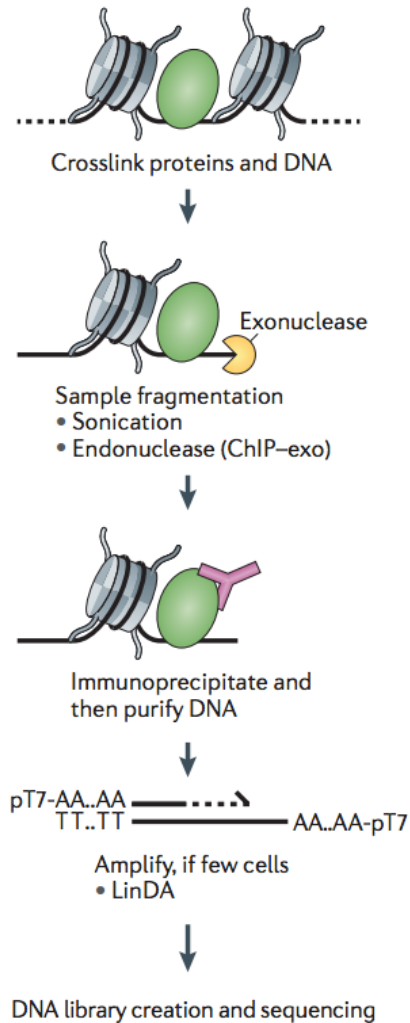
Valouev et al (2008) *Nature Methods*. 5, 829 - 834

ChIP-seq: Histone Modifications

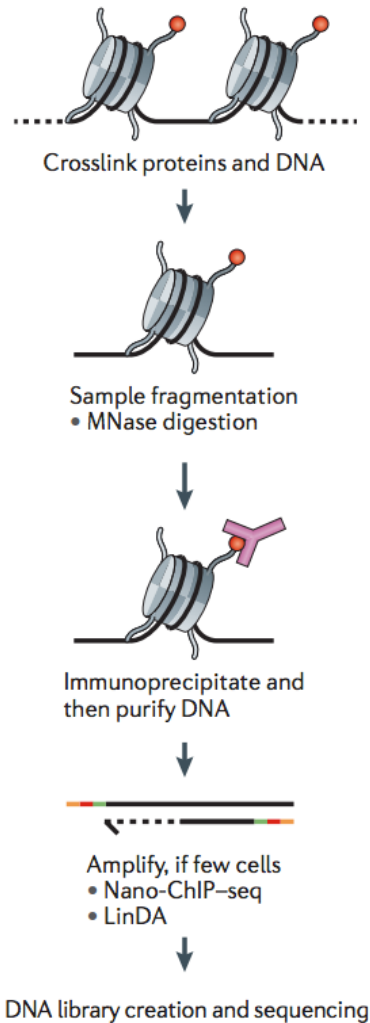


Related Assays

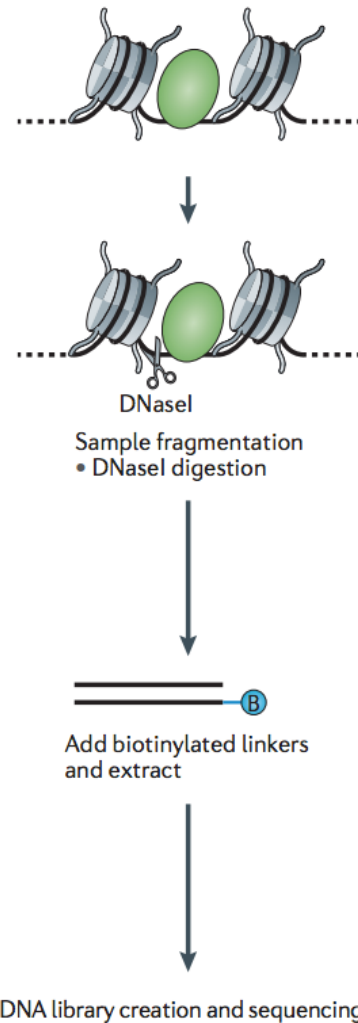
a DNA-binding protein ChIP-seq



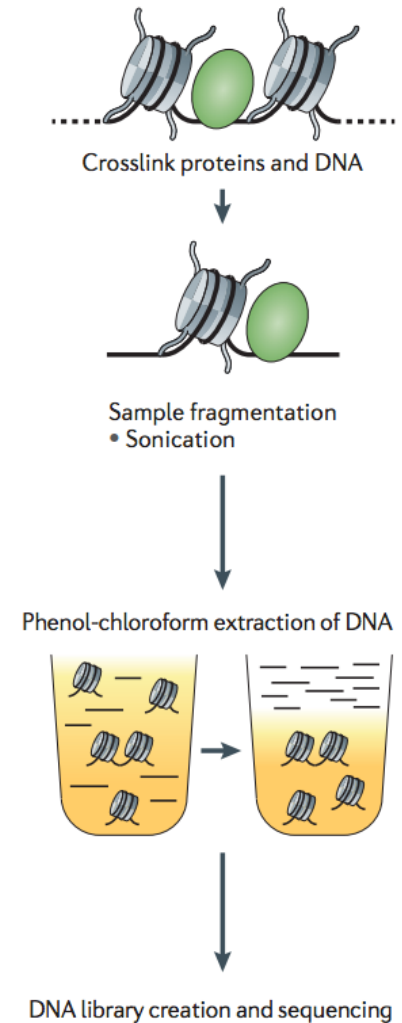
b Histone modification ChIP-seq



c DNase-seq



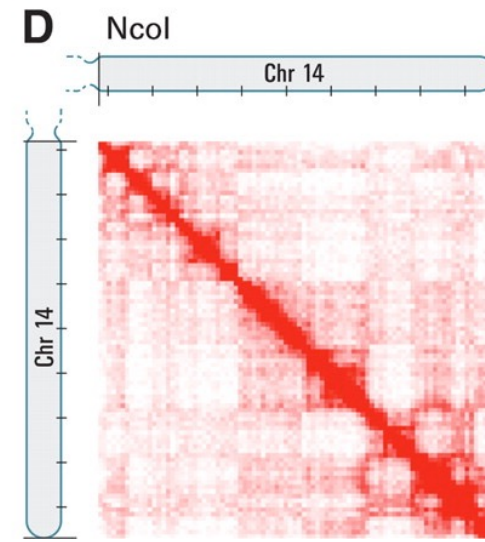
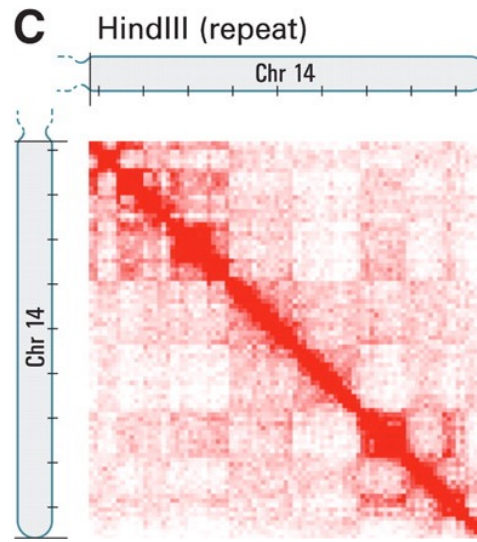
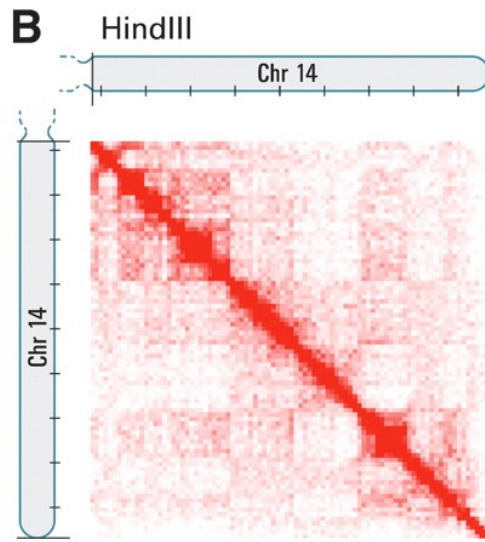
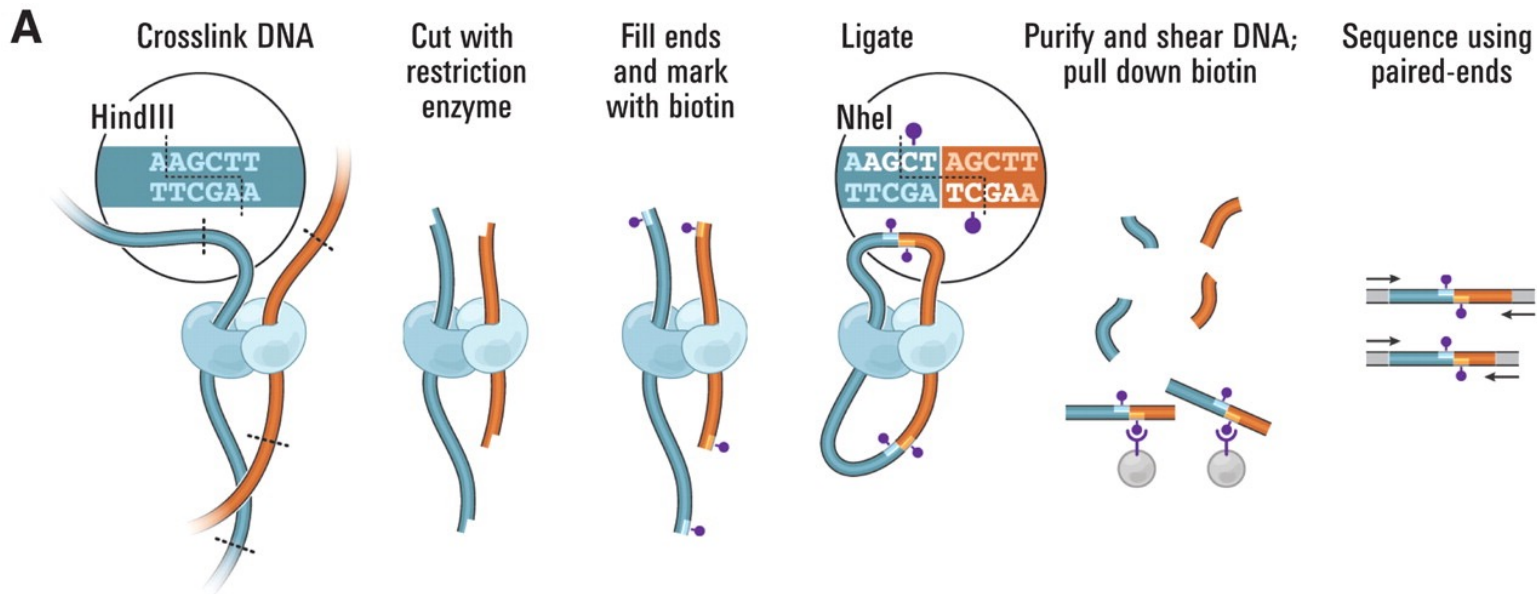
d FAIRE-seq



ChIP-seq and beyond: new and improved methodologies to detect and characterize protein-DNA interactions

Furey (2012) *Nature Reviews Genetics*. 13, 840-852

Hi-C: Mapping the folding of DNA



Comprehensive Mapping of Long-Range Interactions Reveals Folding Principles of the Human Genome

Liberman-Aiden et al. (2009) *Science*. 326 (5950): 289-293

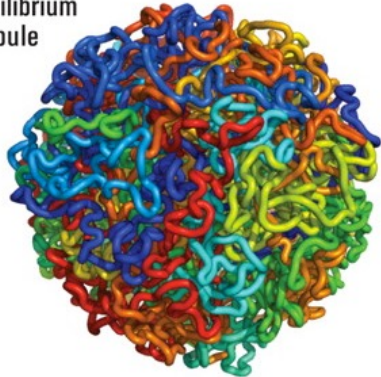
Hi-C: Mapping the folding of DNA

UNFOLDED POLYMER

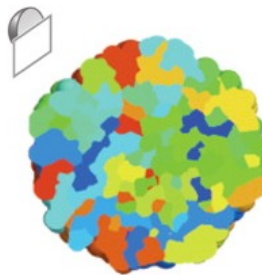


FOLDED POLYMER

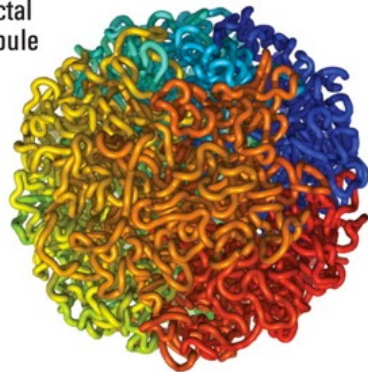
Equilibrium globule



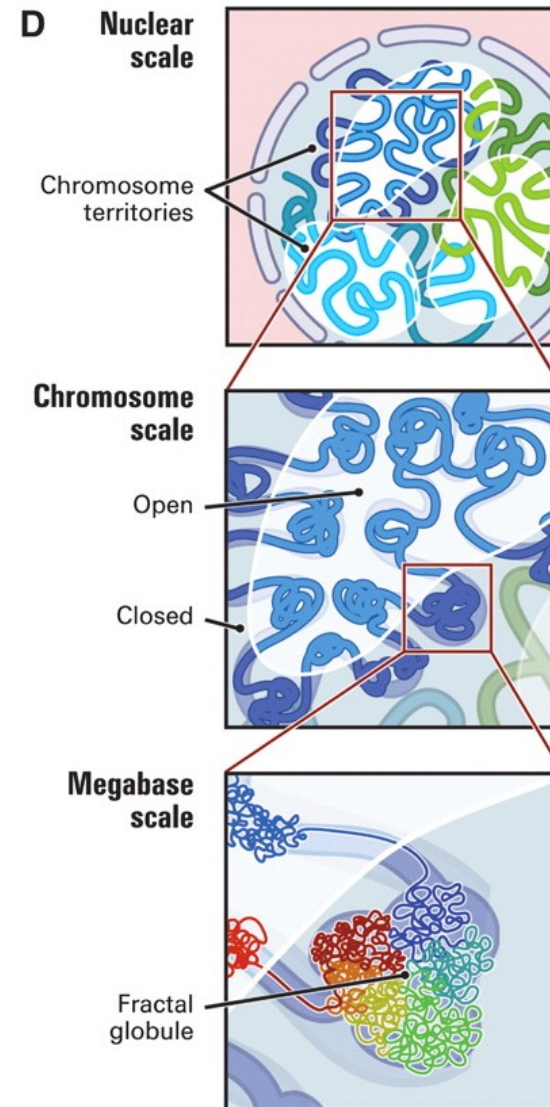
Cross-section view



Fractal globule



Cross-section view



Comprehensive Mapping of Long-Range Interactions Reveals Folding Principles of the Human Genome

Liberman-Aiden et al. (2009) *Science*. 326 (5950): 289-293

Gene Regulation in 3-dimensions

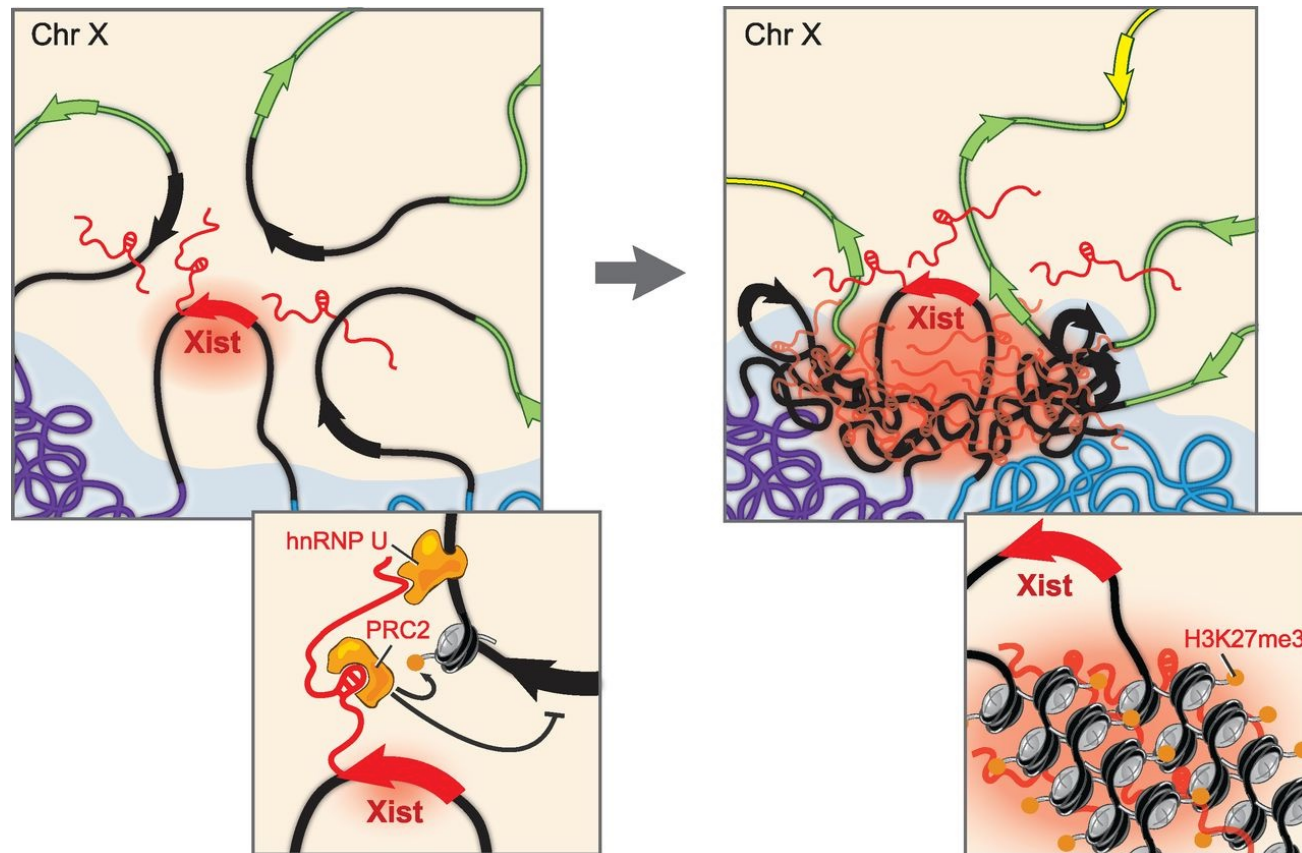
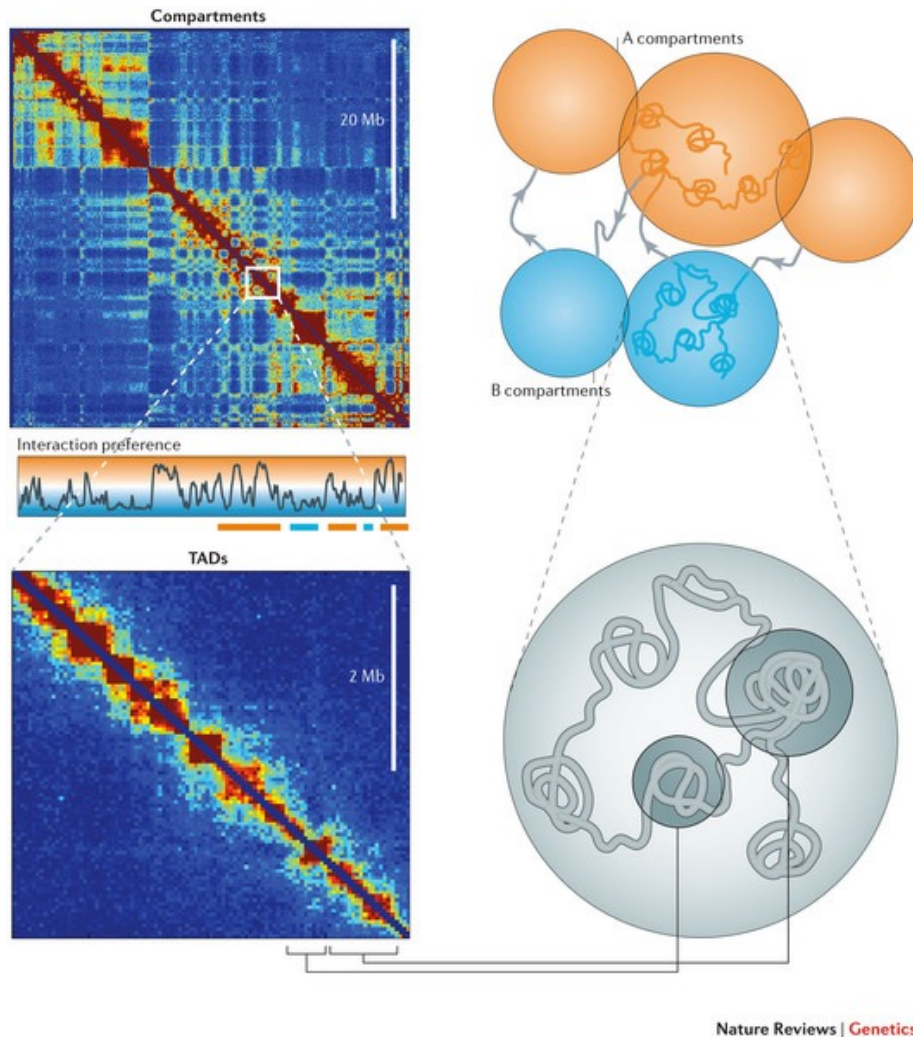


Fig 6. A model for how Xist exploits and alters three-dimensional genome architecture to spread across the X chromosome.

The Xist lncRNA Exploits Three-Dimensional Genome Architecture to Spread Across the X Chromosome
Engreitz et al. (2013) *Science*. 341 (6147)

Genome compartments & TADs



Mammalian genomes have a pattern of interactions that can be approximated by two compartments called A and B

- alternate along chromosomes and have a characteristic size of ~5 Mb each.
- A compartments (orange) preferentially interact with other A compartments; B compartments (blue) associate with other B compartments.
- A compartments are largely euchromatic, transcriptionally active regions.

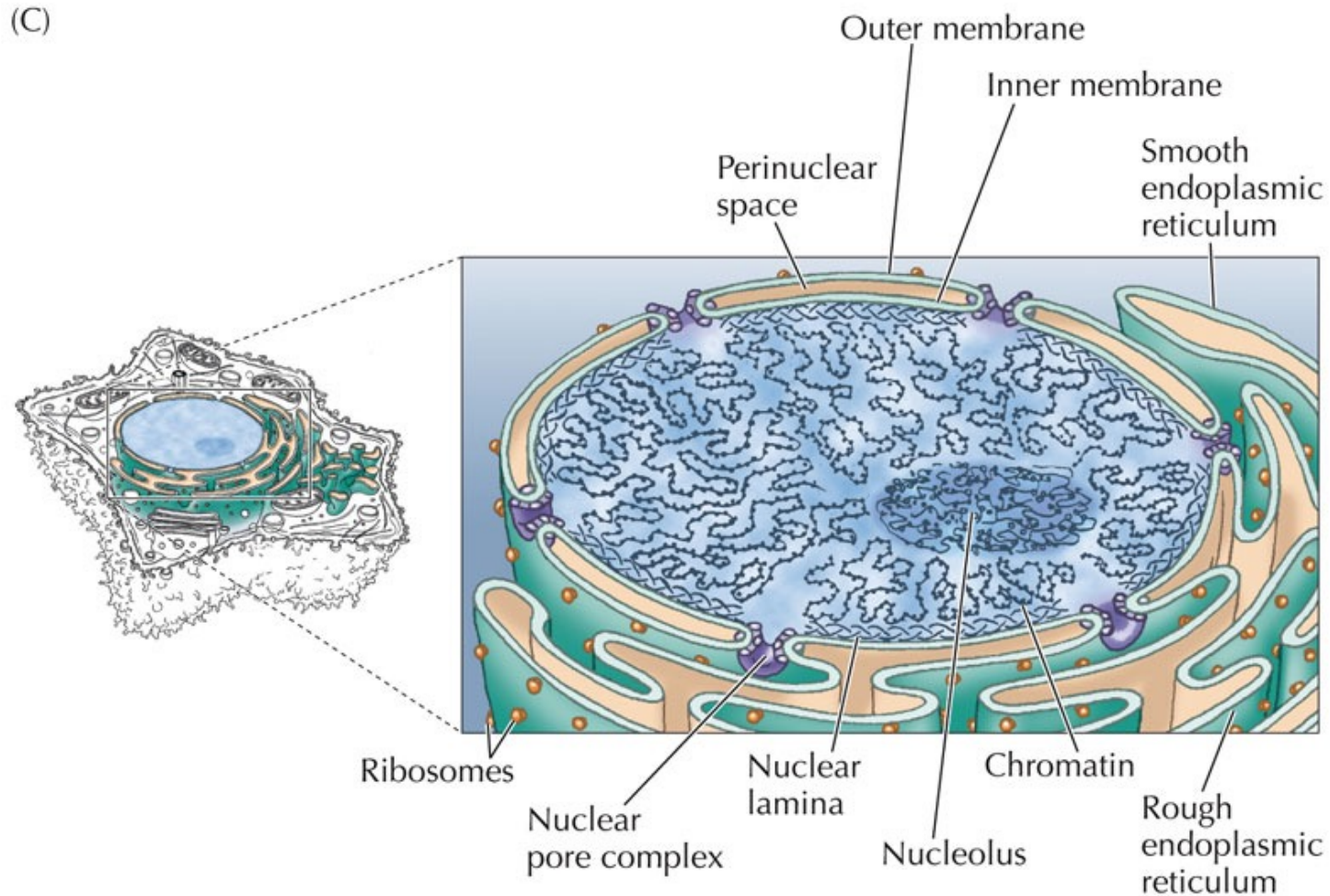
Topologically associating domains (TADs)

- TADs are smaller (~400–500 kb)
- Can be active or inactive, and adjacent TADs are not necessarily of opposite chromatin status.
- TADs are hard-wired features of chromosomes, and groups of adjacent TADs can organize in A and B compartments

Exploring the three-dimensional organization of genomes: interpreting chromatin interaction data

Dekker et al. (2013) *Nature Reviews Genetics* 14, 390–403

“Lamina-Associated Domains are the B compartment”



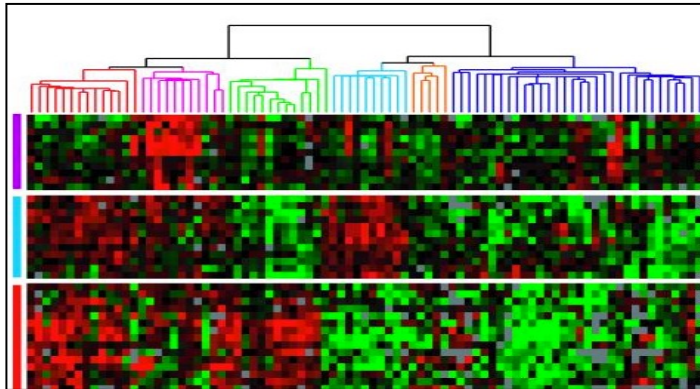
THE CELL, Fourth Edition, Figure 9.1 (Part 3) © 2006 ASM Press and Sinauer Associates, Inc.

Chromosome Conformation Paints Reveal the Role of Lamina Association in Genome Organization and Regulation

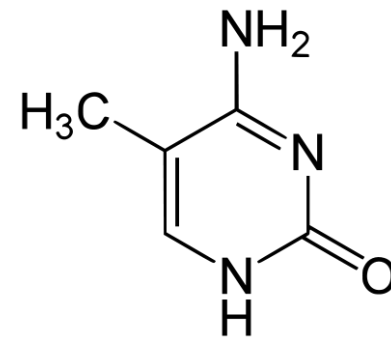
Luperchio et al. (2017) bioRxiv. doi: <https://doi.org/10.1101/122226>

Putting it all together!

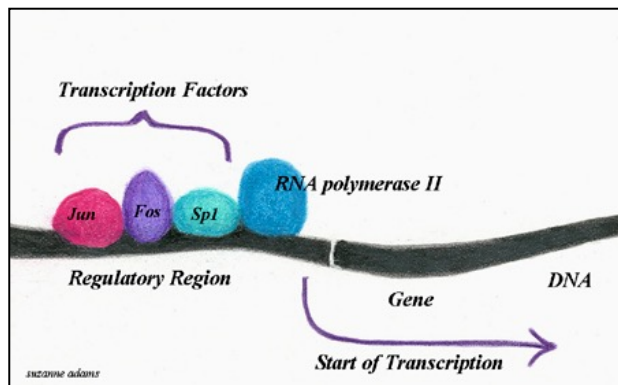
RNA-seq



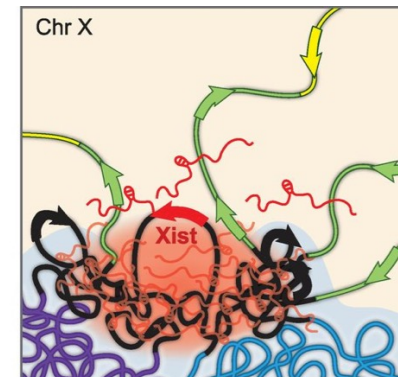
Methyl-seq



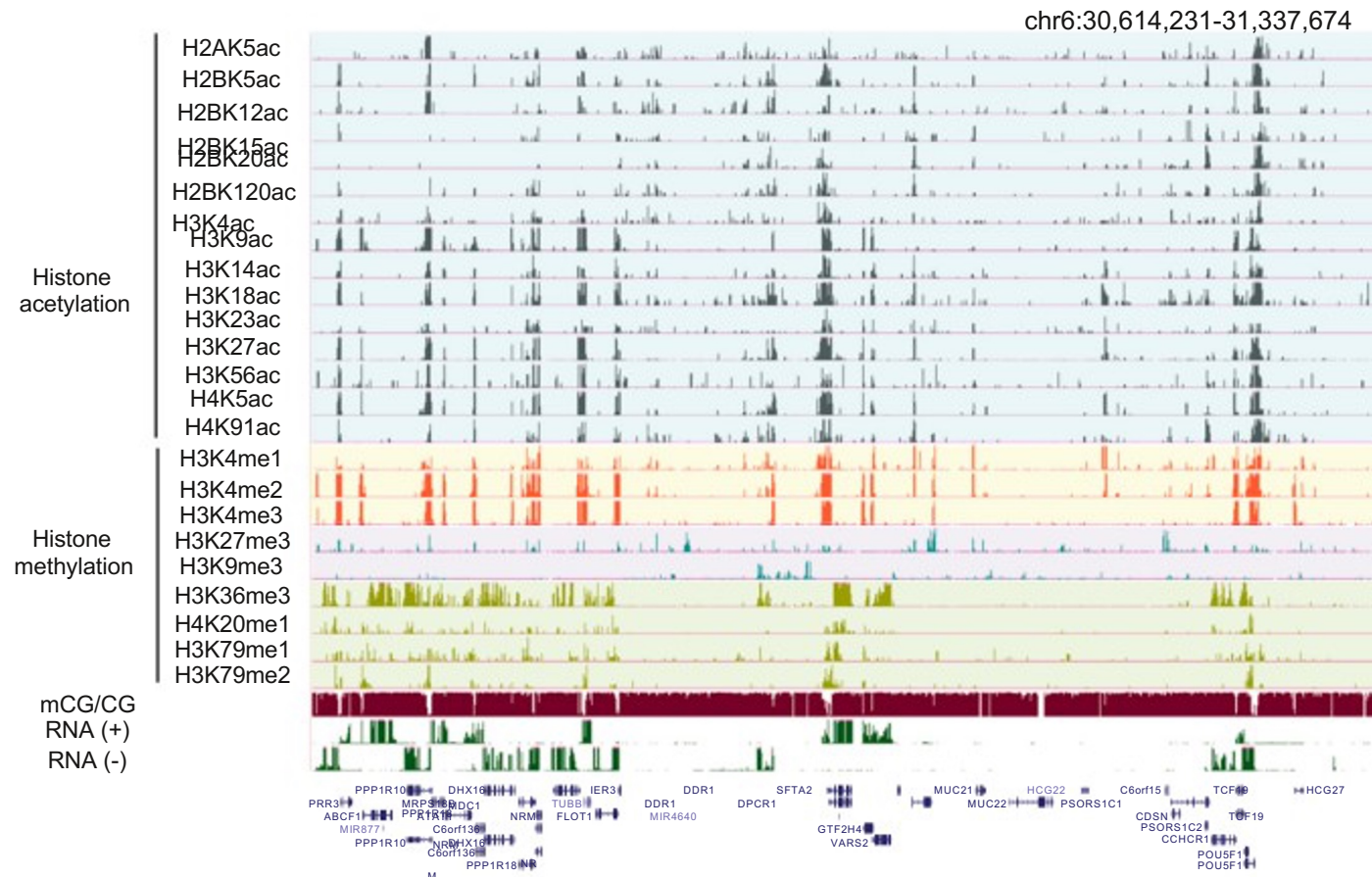
ChIP-seq



Hi-C

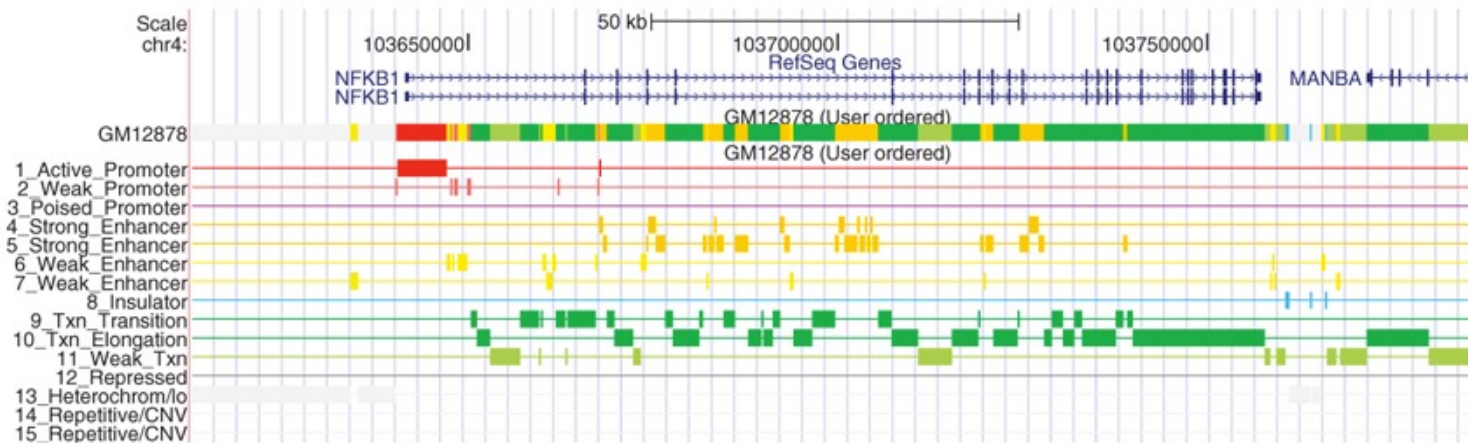


We can call peaks, but...



We need a way to summarize the combinatorial patterns of multiple histone marks into meaningful biological units

ChromHMM



ChromHMM is software for learning and characterizing chromatin states.

- ChromHMM can integrate multiple chromatin datasets such as ChIP-seq data of various histone modifications to discover de novo the major re-occurring combinatorial and spatial patterns of marks.
- ChromHMM is based on a multivariate Hidden Markov Model that explicitly models the presence or absence of each chromatin mark.
- The resulting model can then be used to systematically annotate a genome in one or more cell types.

ChromHMM: automating chromatin-state discovery and characterization

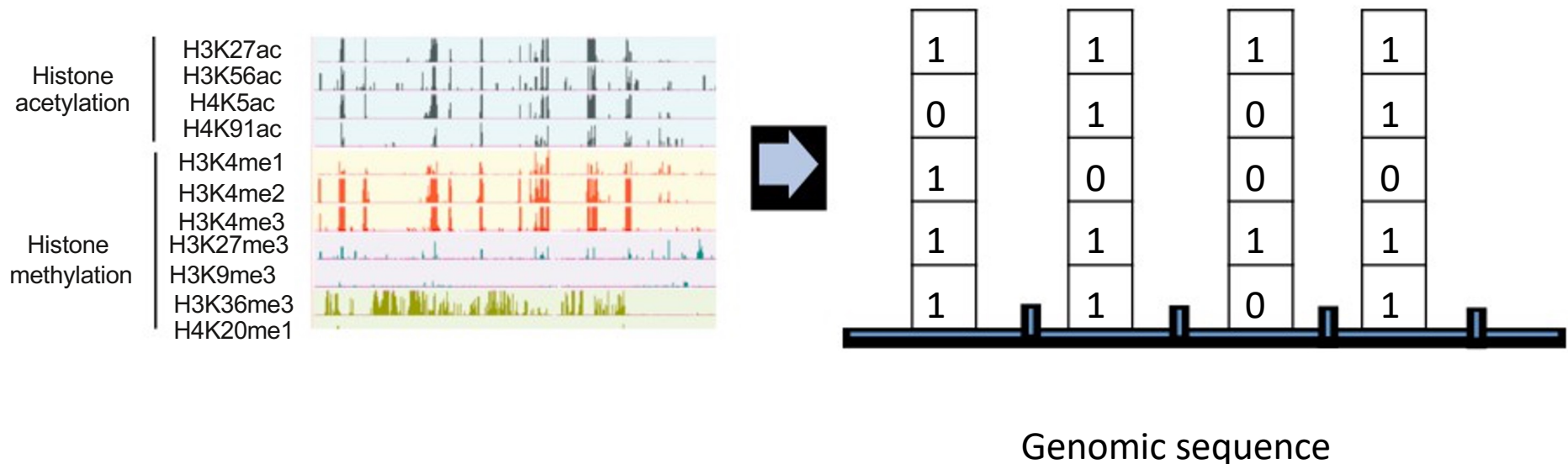
Ernst & Kellis (2012) Nature Methods 9, 215–216. doi:10.1038/nmeth.1906

The Workflow

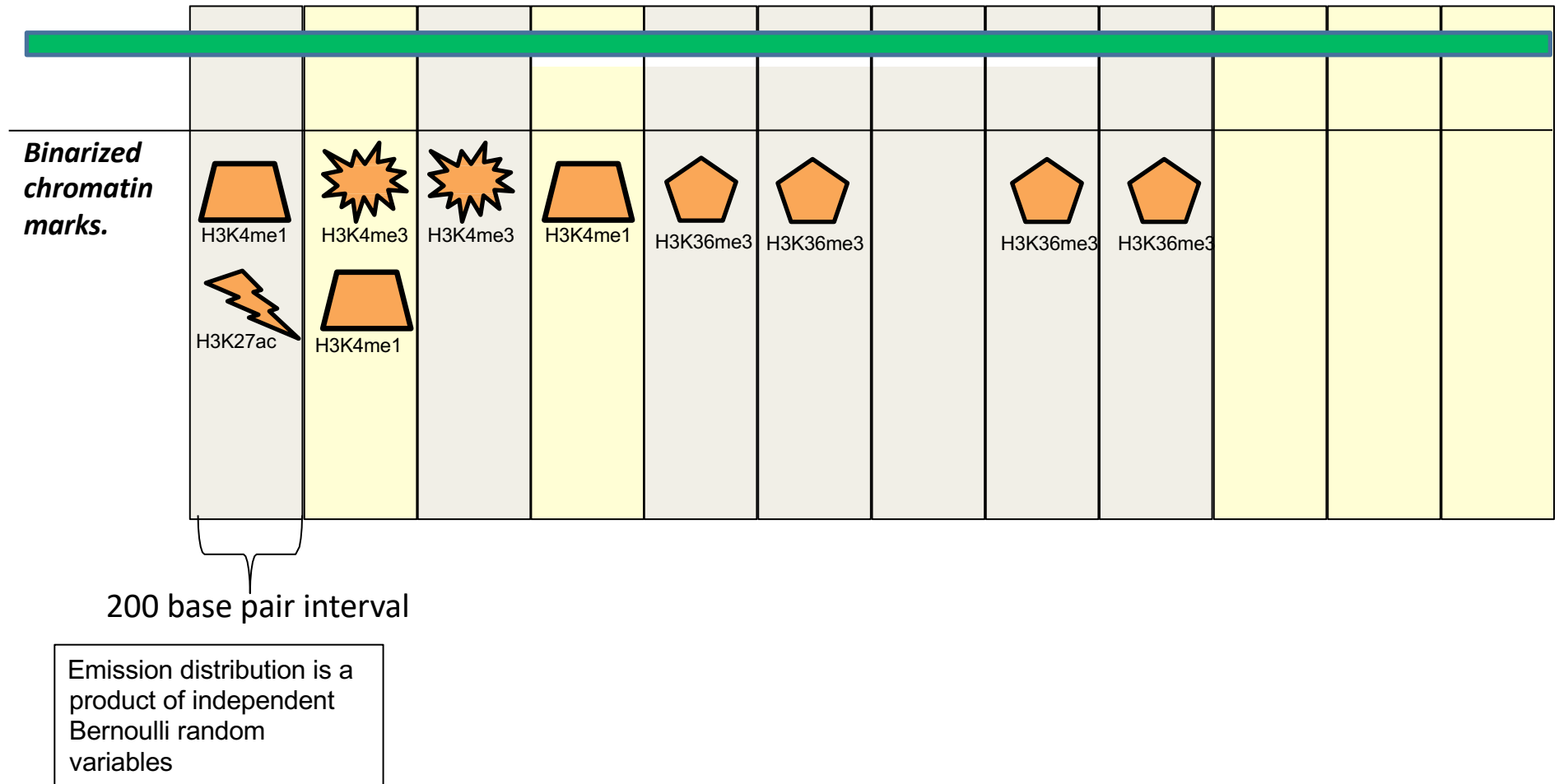
1. Get ChIP-seq raw reads for different histone modifications
2. Align the reads to a reference genome
3. Convert aligned reads in bed format
4. Create Binned and Binarized Tracks
5. Train the model
6. Infer the states
7. Interpretation

Create Binned and Binarized Tracks

- ChromHMM quantify the presence or absence of each mark in bins of fixed size



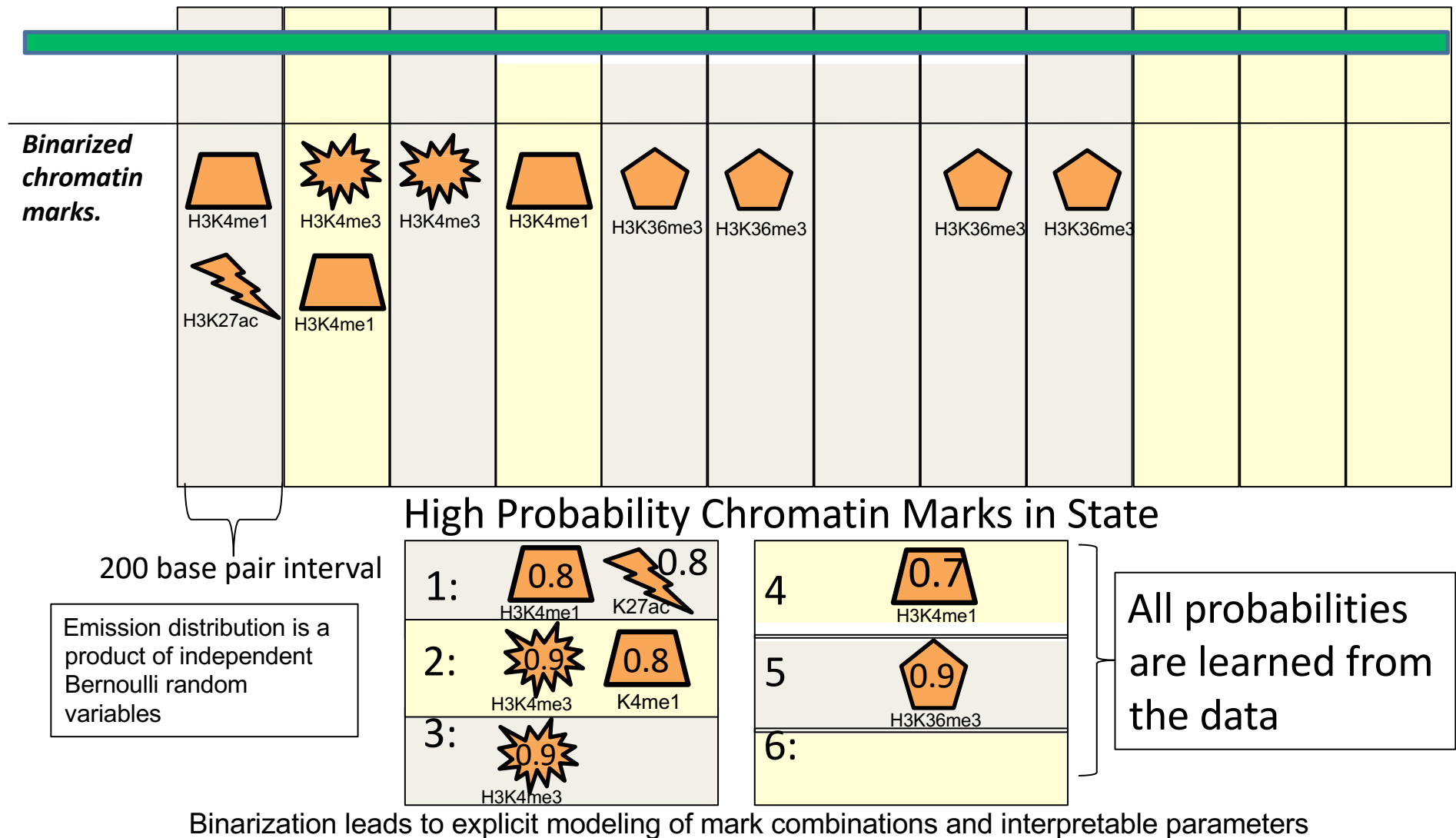
ChromHMM : Multivariate Hidden Markov Model



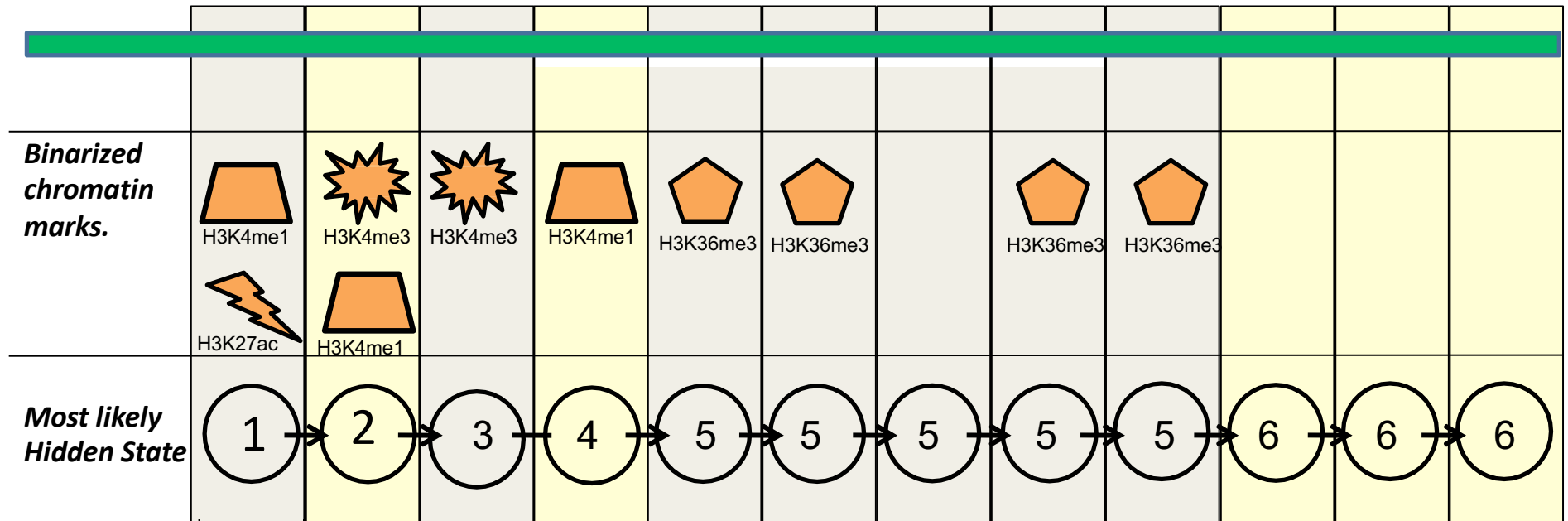
Binarization leads to explicit modeling of mark combinations and interpretable parameters

Ernst and Kellis, Nat Biotech 2010 ; Ernst and Kellis, Nature Methods 2012

ChromHMM : Multivariate Hidden Markov Model



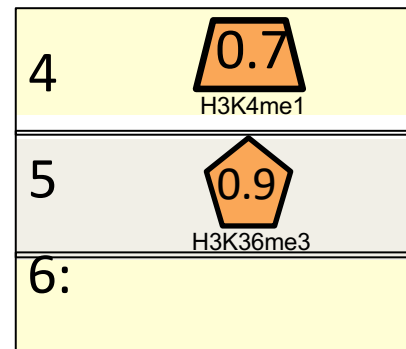
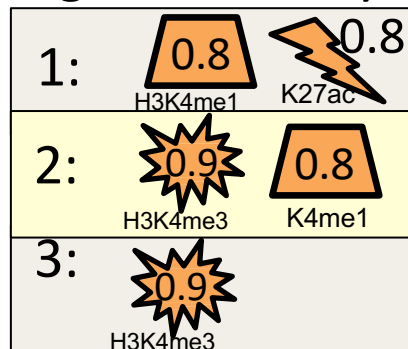
ChromHMM : Multivariate Hidden Markov Model



High Probability Chromatin Marks in State

200 base pair interval

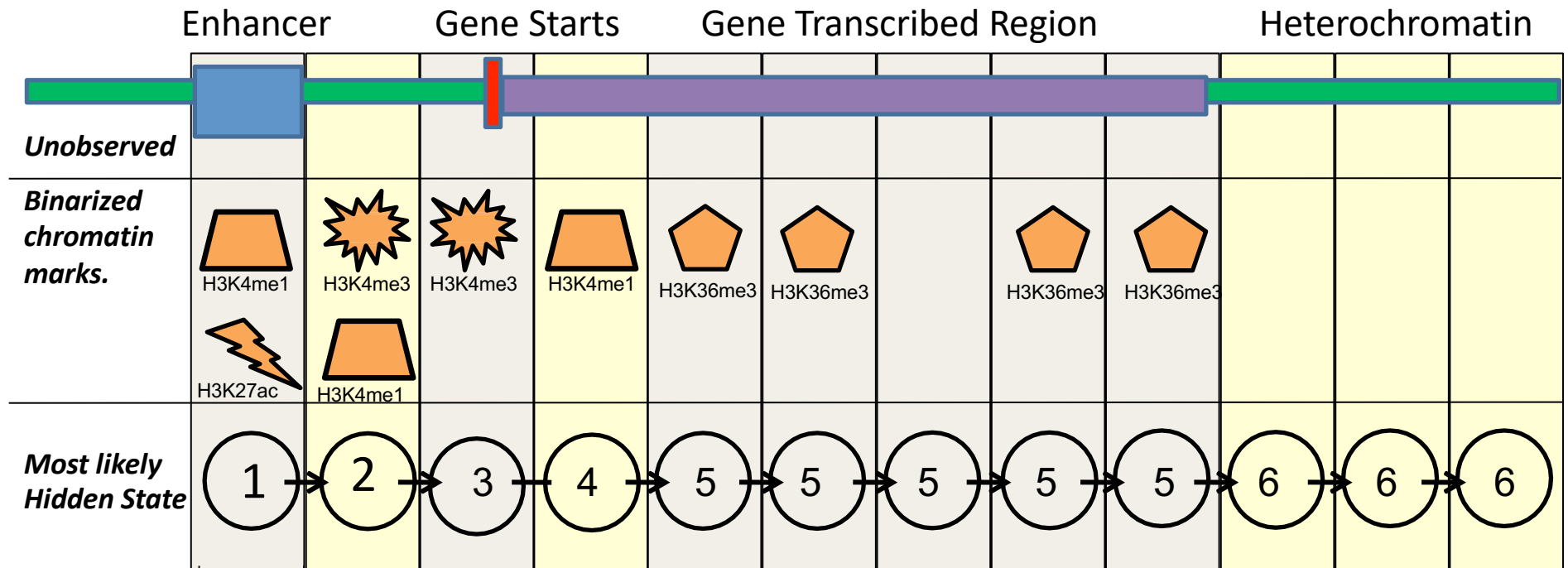
Emission distribution is a product of independent Bernoulli random variables



All probabilities are learned from the data

Binarization leads to explicit modeling of mark combinations and interpretable parameters

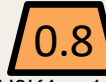
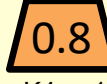
ChromHMM : Multivariate Hidden Markov Model

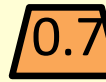


High Probability Chromatin Marks in State

200 base pair interval

Emission distribution is a product of independent Bernoulli random variables

1:	 0.8	 0.8
2:	 0.9	 0.8
3:	 0.9	

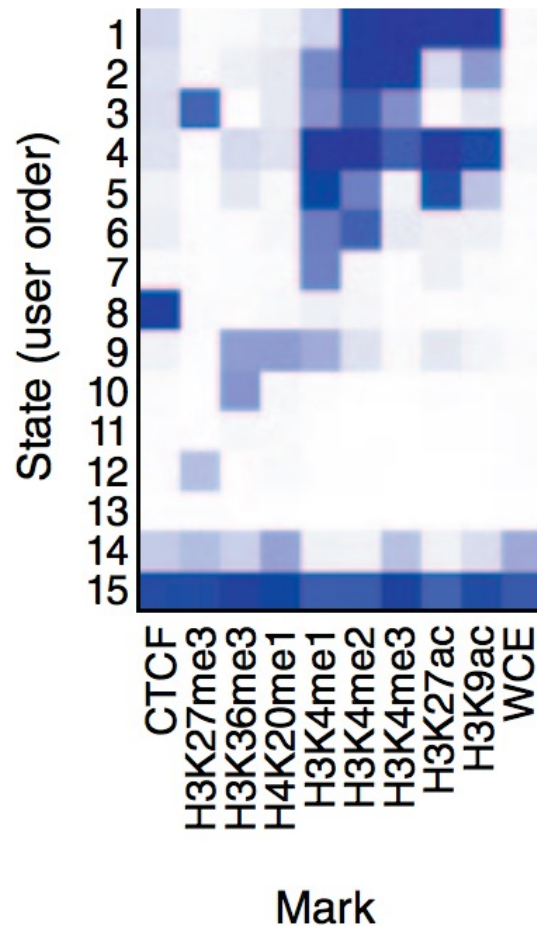
4:	 0.7
5:	 0.9
6:	

All probabilities are learned from the data

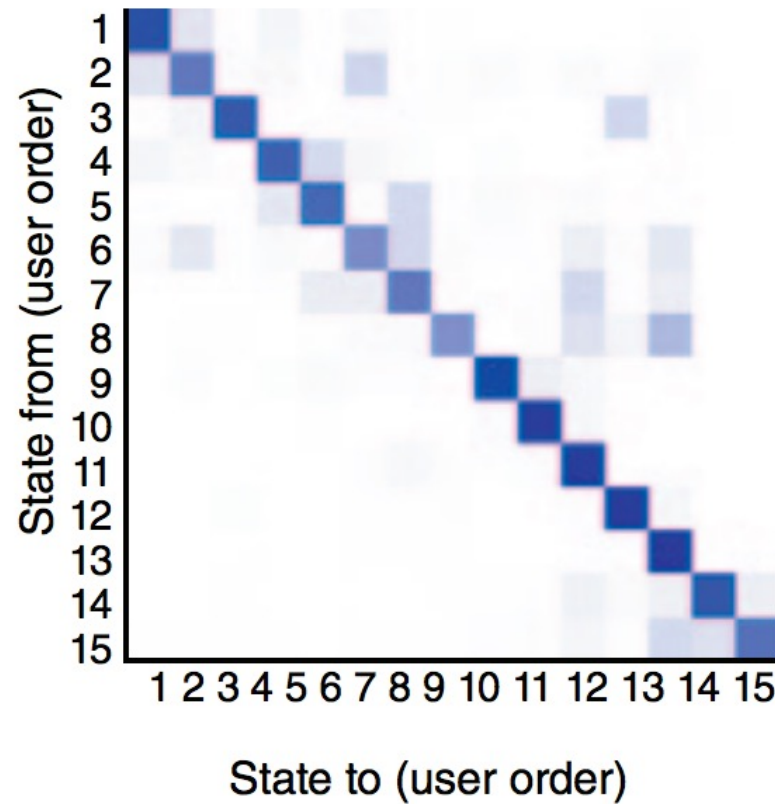
Binarization leads to explicit modeling of mark combinations and interpretable parameters

Transition and Emission Parameters

Emission parameters



Transition parameters



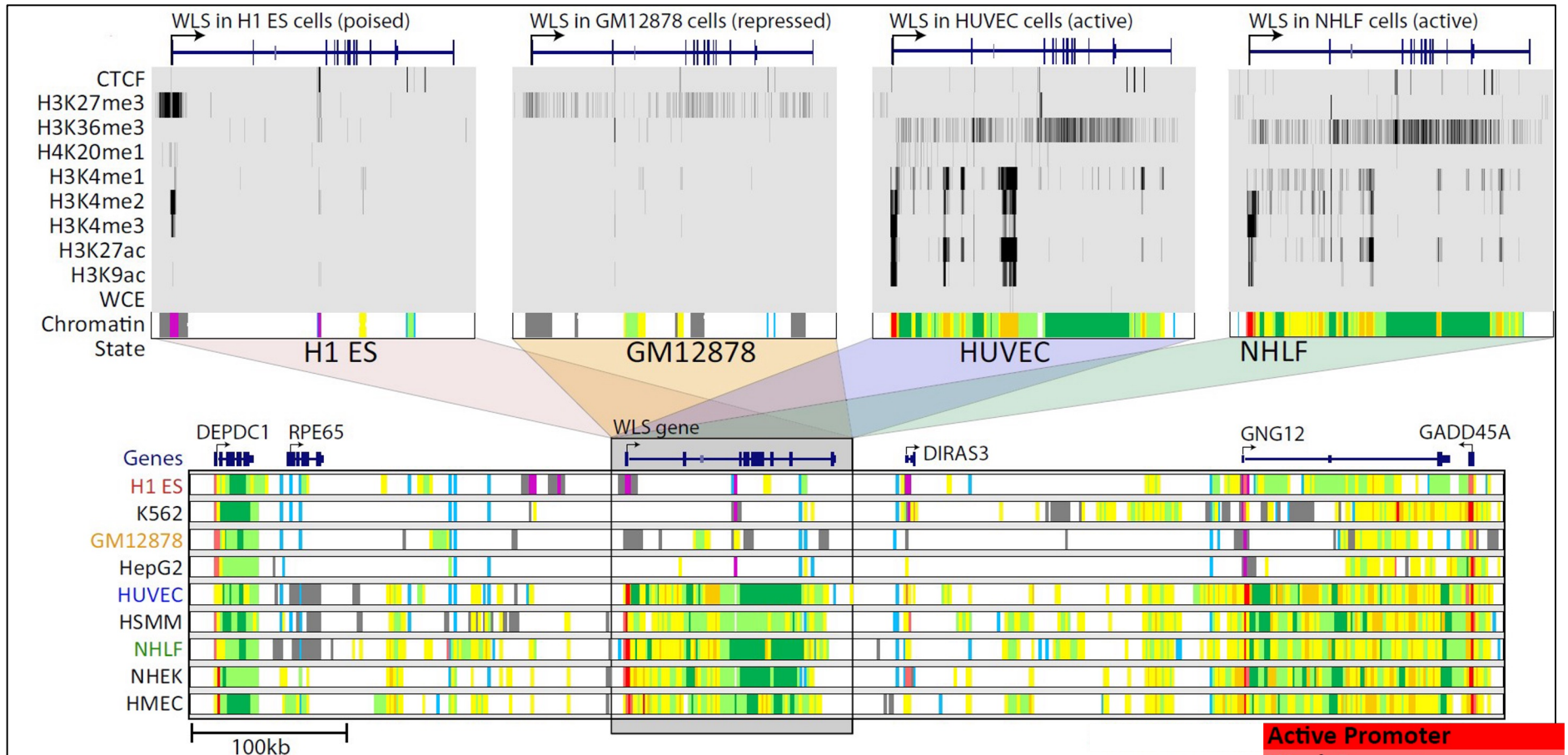
Enriched functional category

The states predicted by the HMM are **statistical** entities (#1 – #15)

The states we want are **biological** entities (Active/Weak/Poised promoter)

Investigate the properties of the statistical entities to label them with biological functions
=> Supervised learning problem ☺

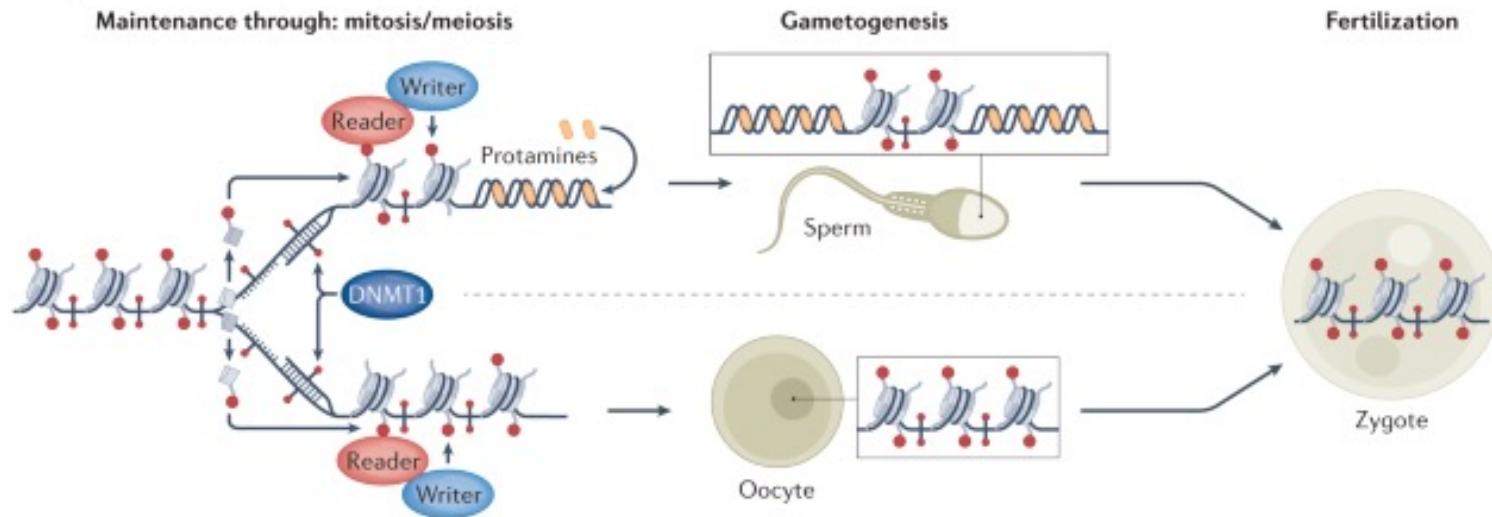
Chromatin states dynamics across nine cell types



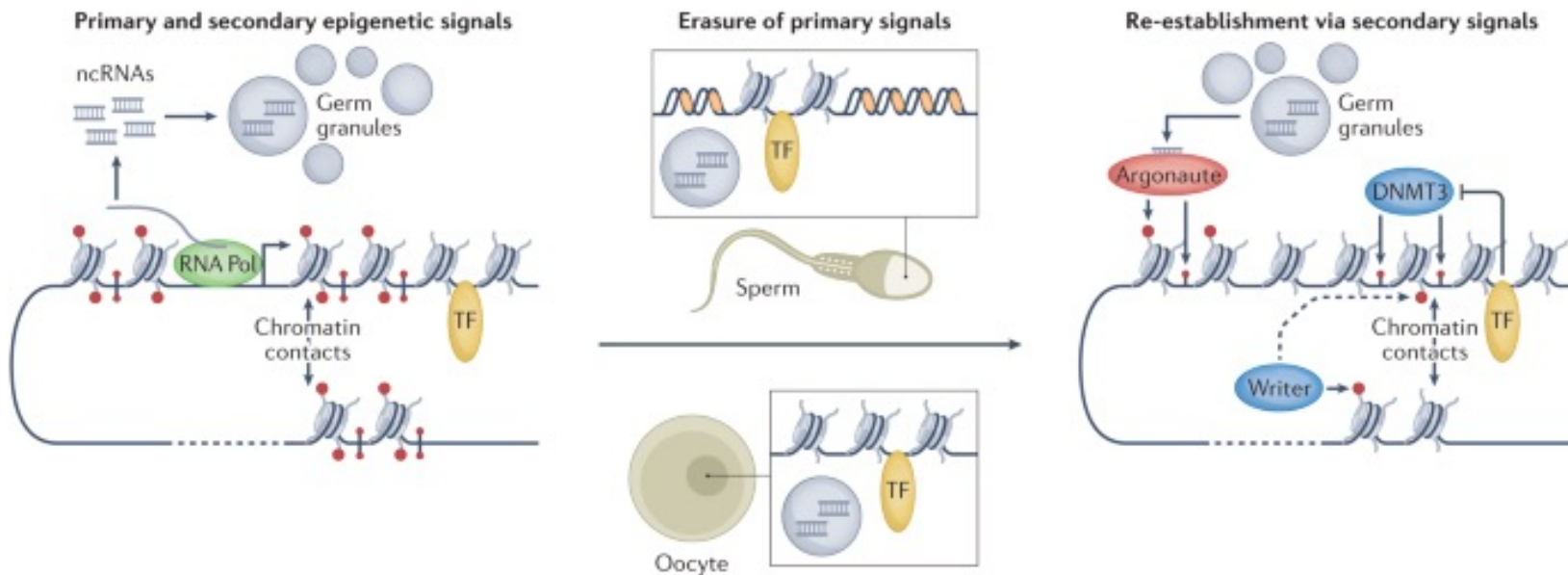
- Single annotation track for each cell type
- Summarize cell-type activity at a glance
- Can study 9-cell activity pattern across ↓

Ernst et al, Nature 2011

a Replicative inheritance

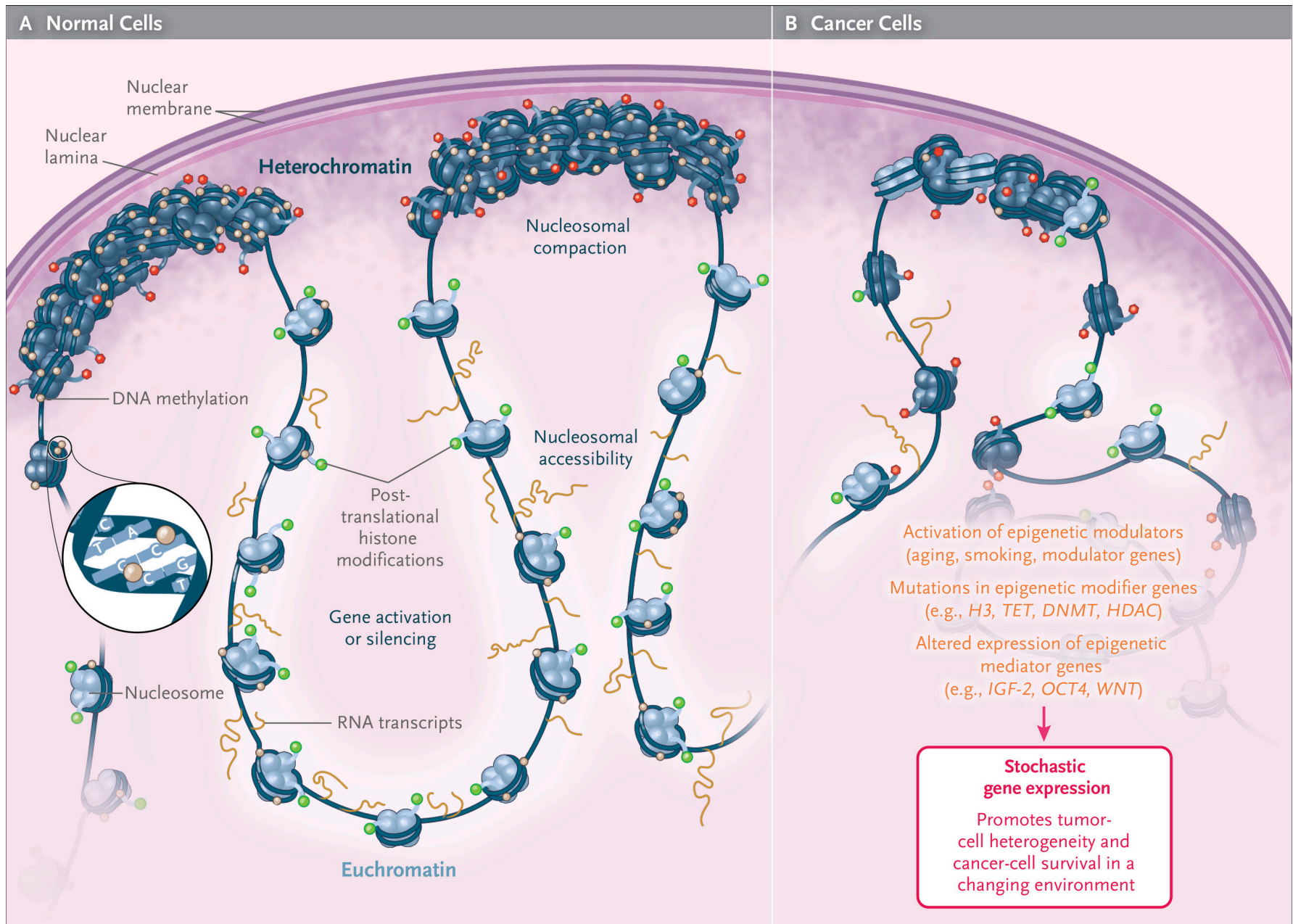


b Reconstructive inheritance



Molecular mechanisms of transgenerational epigenetic inheritance

Fitz-James et al (2022) *Nature Reviews Genetics*. <https://doi.org/10.1038/s41576-021-00438-5>



The Key Role of Epigenetics in Human Disease Prevention and Mitigation

Feinberg (2018) *NEJM*. doi: 10.1056/NEJMr1402513

An integrated encyclopedia of DNA elements in the human genome

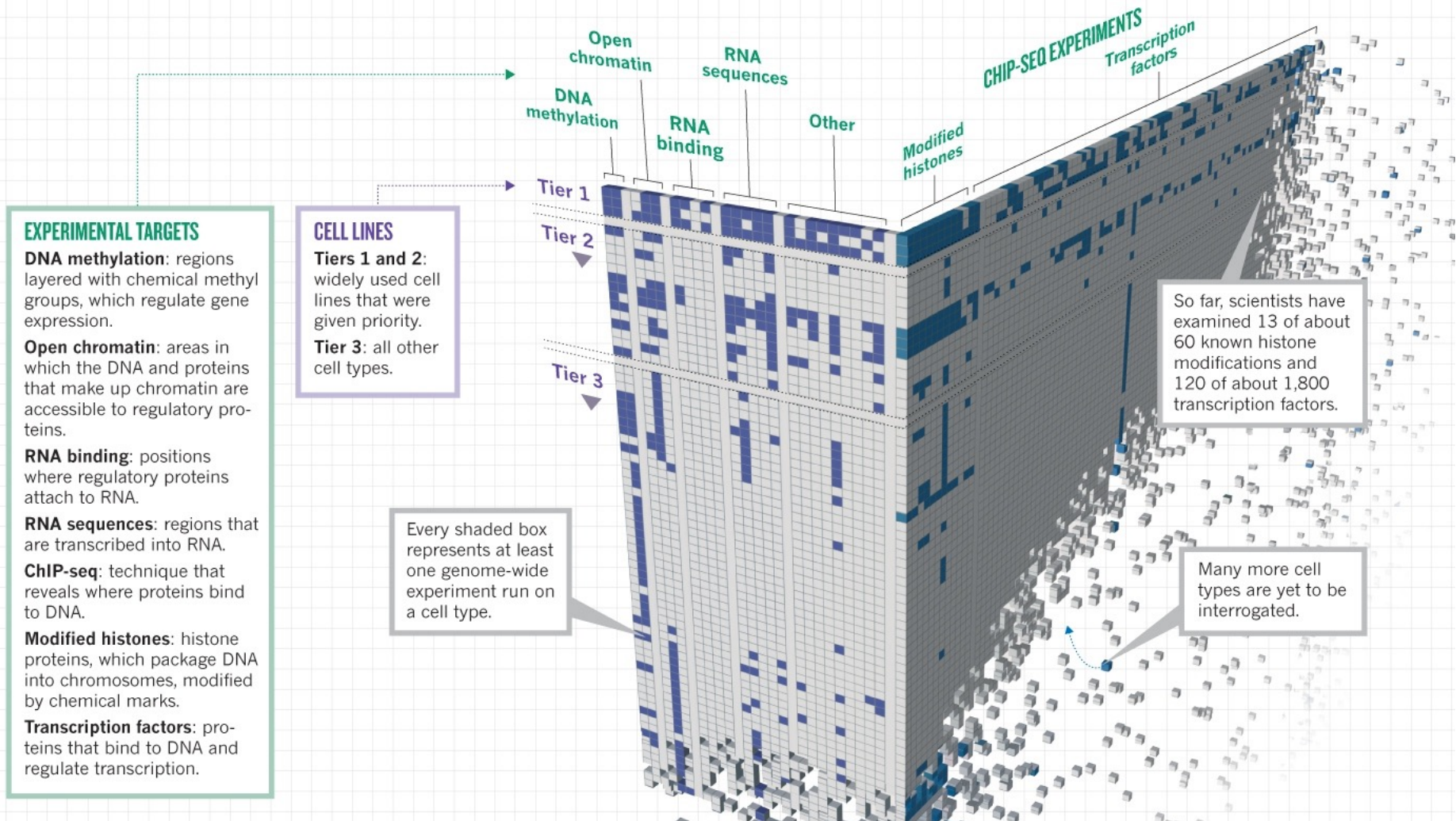
The ENCODE Project Consortium*

The human genome encodes the blueprint of life, but the function of the vast majority of its nearly three billion bases is unknown. The Encyclopedia of DNA Elements (ENCODE) project has systematically mapped regions of transcription, transcription factor association, chromatin structure and histone modification. These data enabled us to assign biochemical functions for 80% of the genome, in particular outside of the well-studied protein-coding regions. Many discovered candidate regulatory elements are physically associated with one another and with expressed genes, providing new insights into the mechanisms of gene regulation. The newly identified elements also show a statistical correspondence to sequence variants linked to human disease, and can thereby guide interpretation of this variation. Overall, the project provides new insights into the organization and regulation of our genes and genome, and is an expansive resource of functional annotations for biomedical research.

ENCODE Data Sets

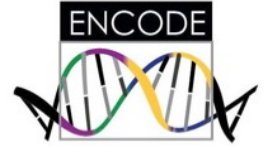
MAKING A GENOME MANUAL

Scientists in the Encyclopedia of DNA Elements Consortium have applied 24 experiment types (across) to more than 150 cell lines (down) to assign functions to as many DNA regions as possible — but the project is still far from complete.



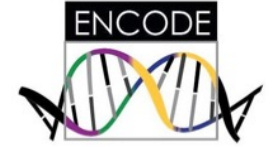
1,640 data sets total over 147 different cell types

Major Findings



1. *The vast majority (80.4%) of the human genome participates in at least one biochemical RNA- and/or chromatin-associated event in at least one cell type.*
2. *Primate-specific elements as well as elements without detectable mammalian constraint show, in aggregate, evidence of negative selection; thus, some of them are expected to be functional.*
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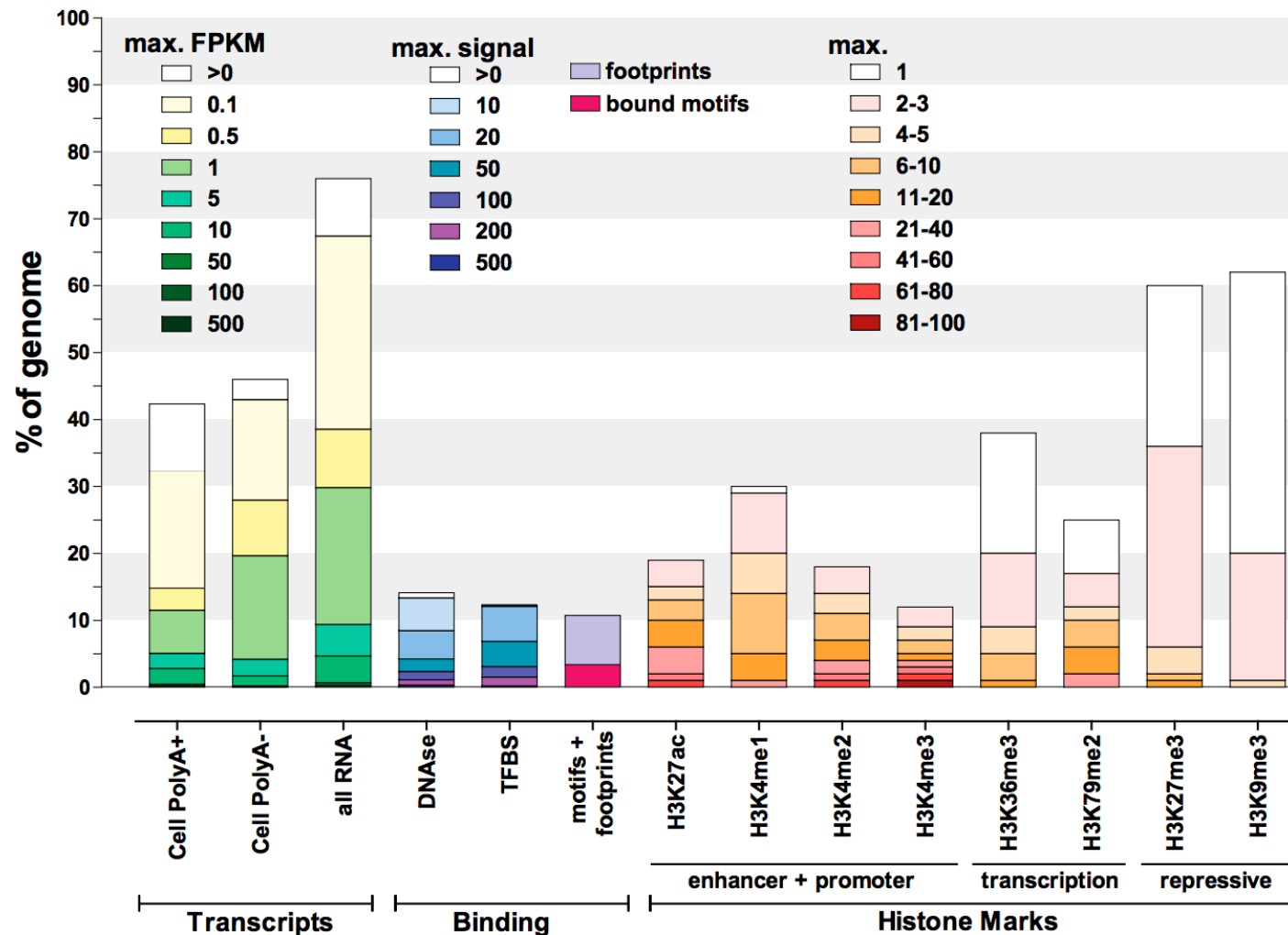
Summary of ENCODE elements

“Accounting for all these elements, a surprisingly large amount of the human genome, 80.4%, is covered by at least one ENCODE-identified element”

- 62% transcribed
- 56% enriched for histone marks
- 15% open chromatin
- 8% TF binding
- 19% At least one DHS or TF Chip-seq peak
- 4% TF binding site motif
- (Note protein coding genes comprise ~2.94% of the genome)

*“Given that the ENCODE project did not assay all cell types, or all transcription factors, and in particular has sampled few specialized or developmentally restricted cell lineages, **these proportions must be underestimates of the total amount of functional bases.**”*

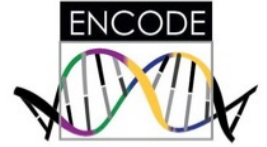
Pervasive Transcription and Regulation



Defining functional DNA elements in the human genome

Kellis et al (2014). *PNAS* 6131–6138, doi: 10.1073/pnas.1318948111

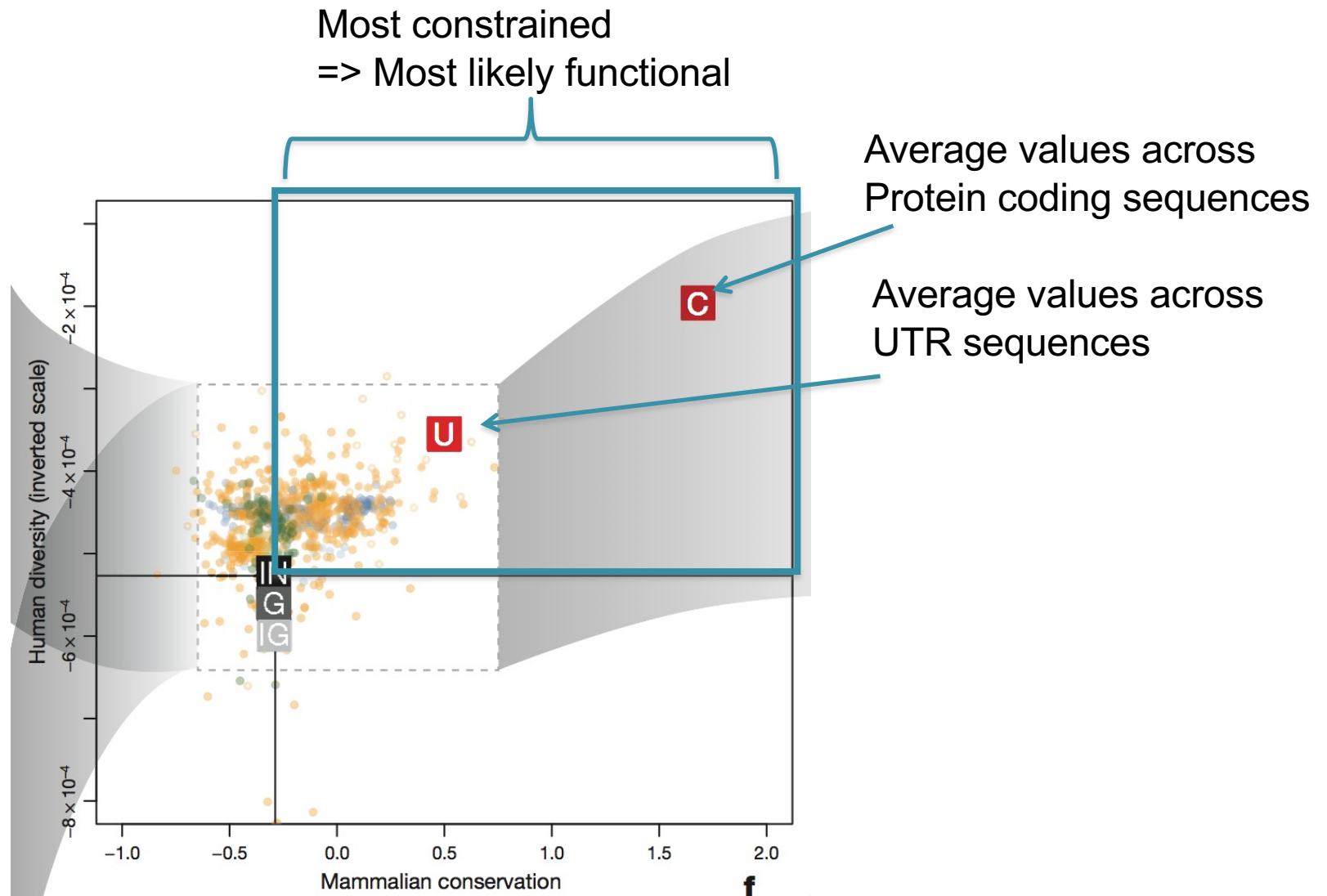
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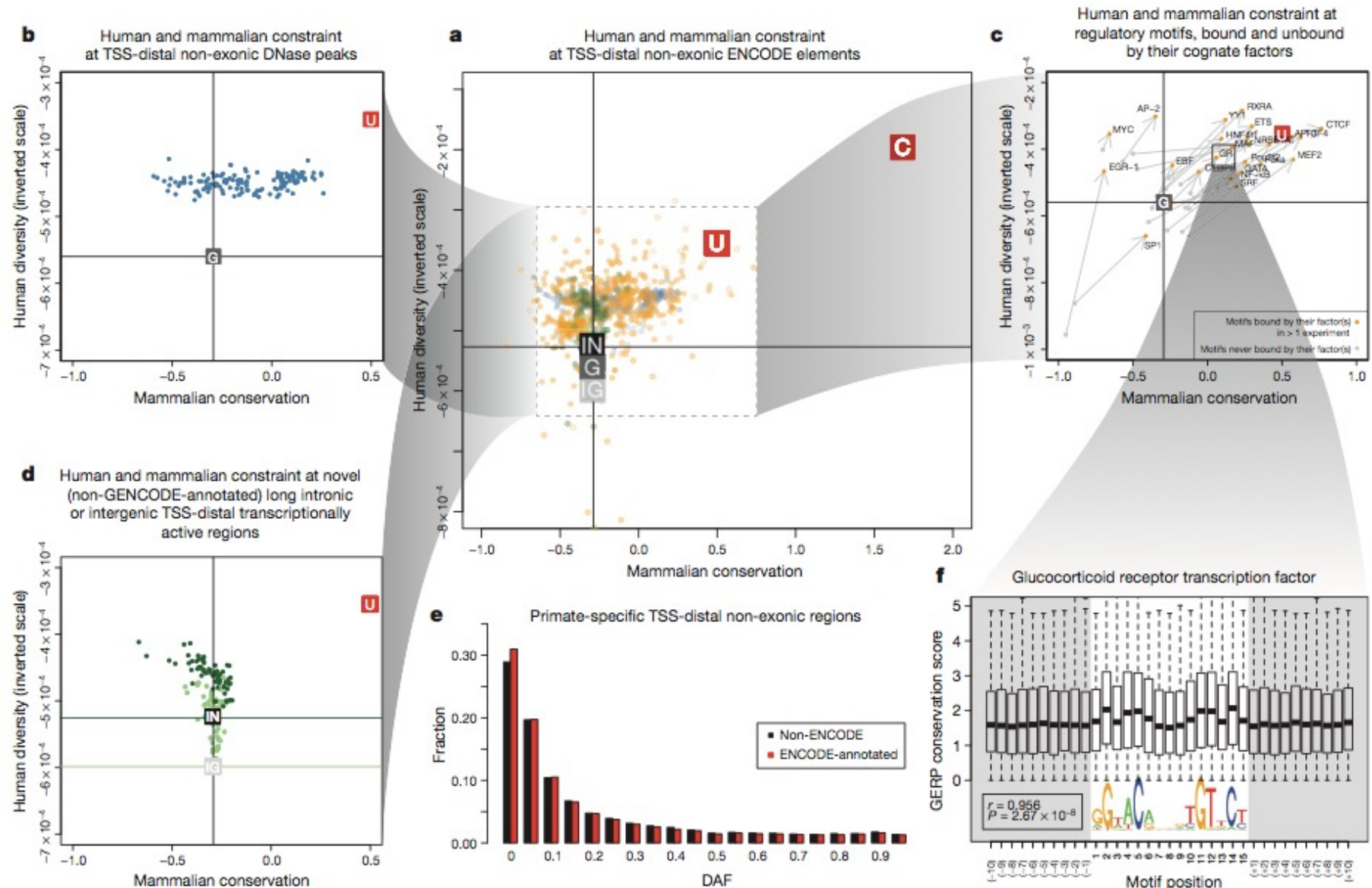
Impact and Evidence of Selection

For a given ENCODE region, how much conservation do we see across modern humans (1000 genomes project)

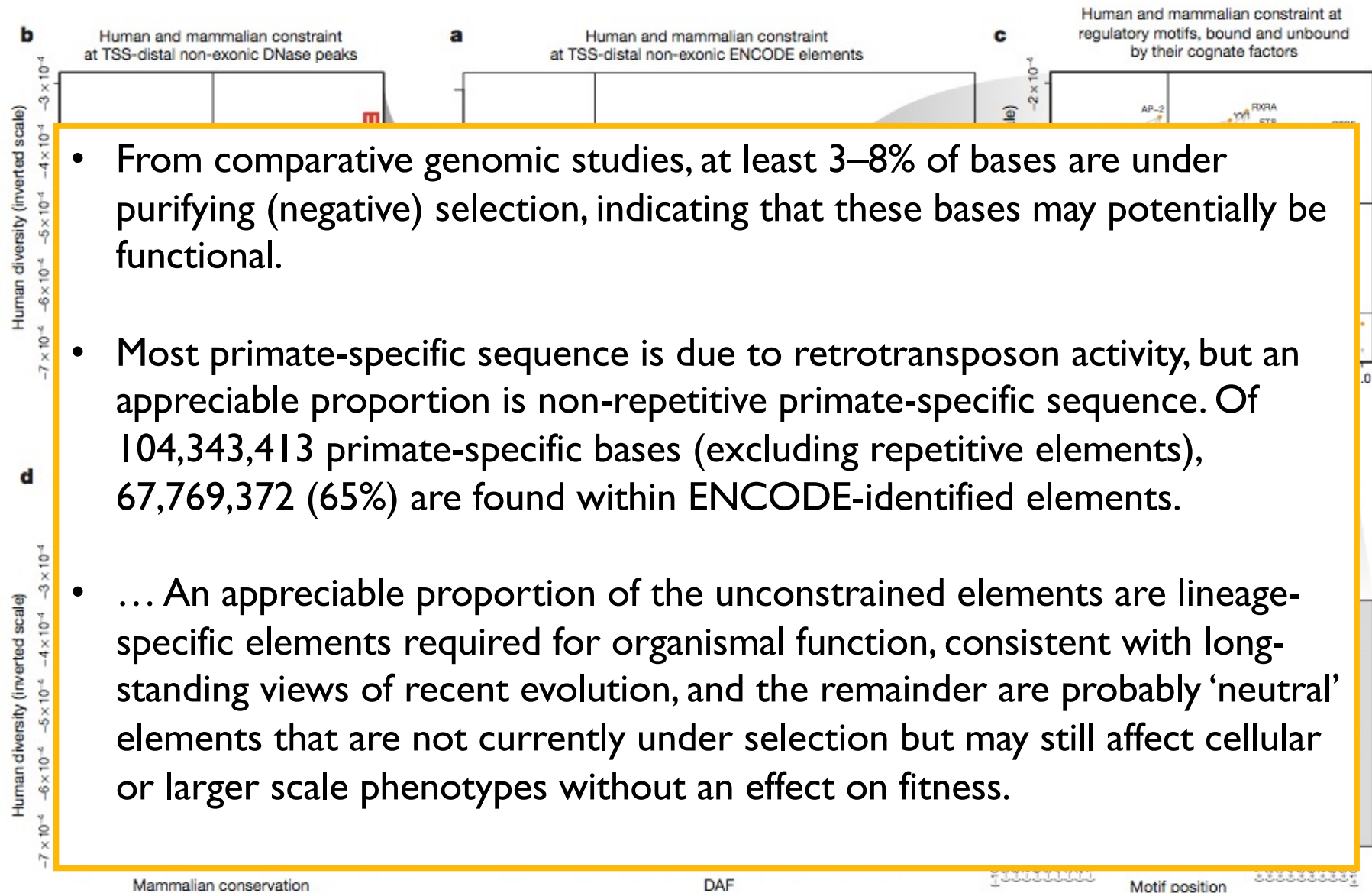


For a given ENCODE region, how much conservation do we see across 24 sequenced mammalian genomes?

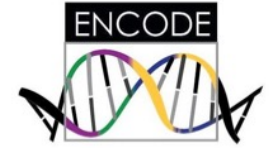
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Impact and Evidence of Selection



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Signal Integration

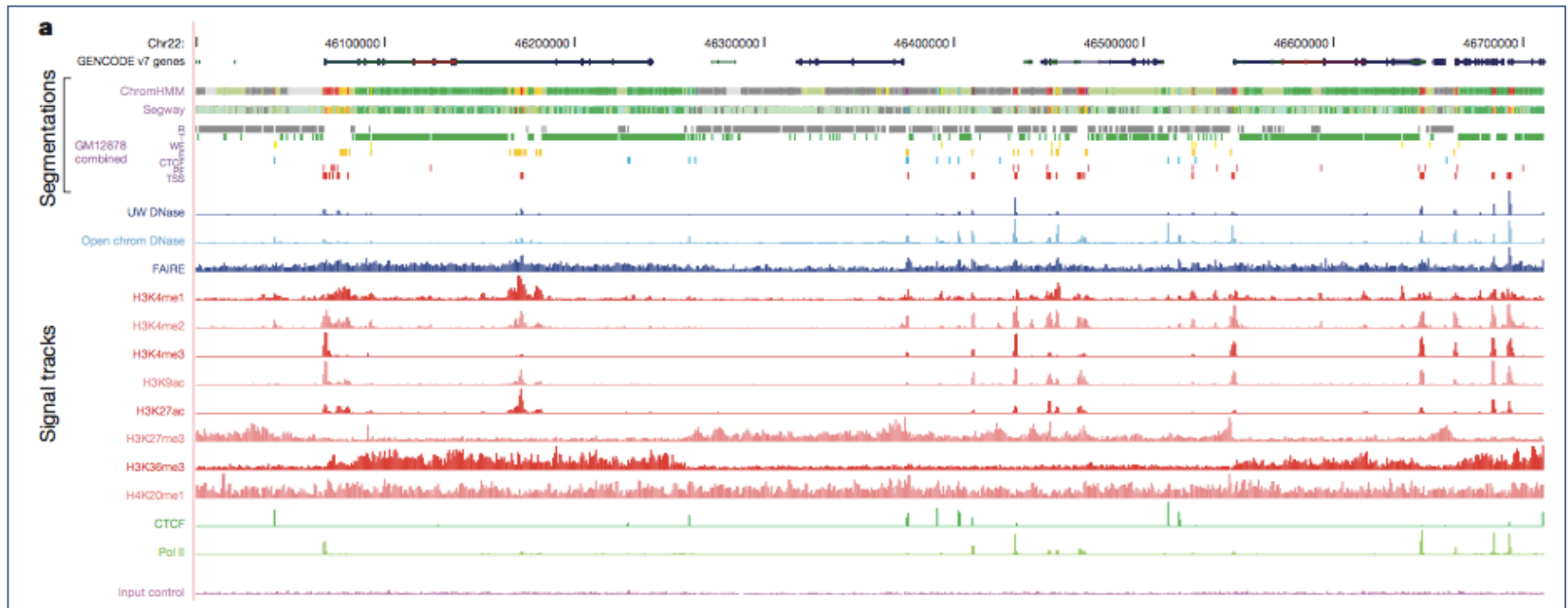
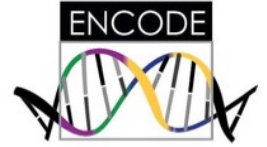


Table 3 | Summary of the combined state types

Label	Description	Details*	Colour
CTCF	CTCF-enriched element	Sites of CTCF signal lacking histone modifications, often associated with open chromatin. Many probably have a function in insulator assays, but because of the multifunctional nature of CTCF, we are conservative in our description. Also enriched for the cohesin components RAD21 and SMC3; CTCF is known to recruit the cohesin complex.	Turquoise
E	Predicted enhancer	Regions of open chromatin associated with H3K4me1 signal. Enriched for other enhancer-associated marks, including transcription factors known to act at enhancers. In enhancer assays, many of these (>50%) function as enhancers. A more conservative alternative would be cis-regulatory regions. Enriched for sites for the proteins encoded by <i>EP300</i> , <i>FOS</i> , <i>FOSL1</i> , <i>GATA2</i> , <i>HDAC8</i> , <i>JUNB</i> , <i>JUND</i> , <i>NFE2</i> , <i>SMARCA4</i> , <i>SMARCB1</i> , <i>SIRT6</i> and <i>TAL1</i> genes in K562 cells. Have nuclear and whole-cell RNA signal, particularly poly(A) ⁺ fraction.	Orange
PF	Predicted promoter flanking region	Regions that generally surround TSS segments (see below).	Light red
R	Predicted repressed or low-activity region	This is a merged state that includes H3K27me3 polycomb-enriched regions, along with regions that are silent in terms of observed signal for the input assays to the segmentations (low or no signal). They may have other signals (for example, RNA, not in the segmentation input data). Enriched for sites for the proteins encoded by <i>REST</i> and some other factors (for example, proteins encoded by <i>BRF2</i> , <i>CEBPB</i> , <i>MAFK</i> , <i>TRIM28</i> , <i>ZNF274</i> and <i>SETDB1</i> genes in K562 cells).	Grey
TSS	Predicted promoter region including TSS	Found close to or overlapping GENCODE TSS sites. High precision/recall for TSSs. Enriched for H3K4me3. Sites of open chromatin. Enriched for transcription factors known to act close to promoters and polymerases Pol II and Pol III. Short RNAs are most enriched in these segments.	Bright red
T	Predicted transcribed region	Overlap gene bodies with H3K36me3 transcriptional elongation signal. Enriched for phosphorylated form of Pol II signal (elongating polymerase) and poly(A) ⁺ RNA, especially cytoplasmic.	Dark green
WE	Predicted weak enhancer or open chromatin cis-regulatory element	Similar to the E state, but weaker signals and weaker enrichments.	Yellow

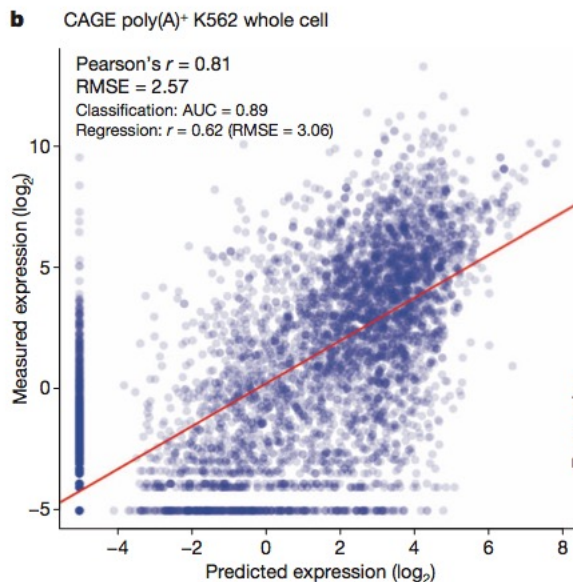
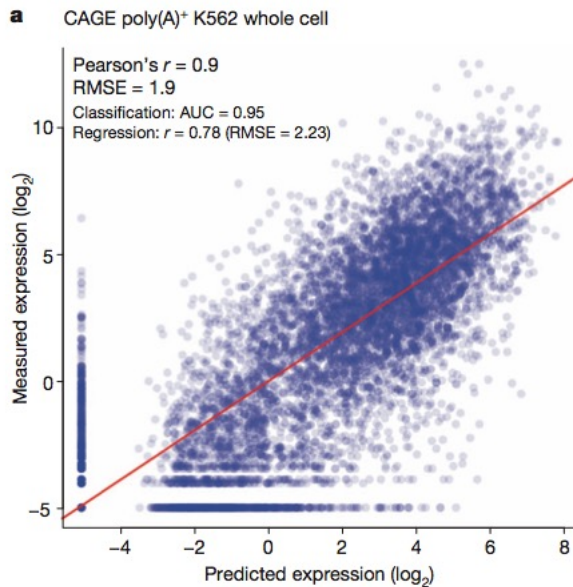
- Use ChromHMM and Segway to Summarize the individual assays into 7 functional/regulatory states

Major Findings



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Expression Modeling

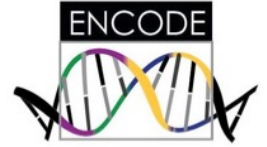


- Developed predictive models to explore the interaction between histone modifications and transcription factor binding towards level of transcription
- The best models had two components: an initial classification component (on/off) and a second quantitative model component
- Together, these correlation models indicate both that a limited set of chromatin marks are sufficient to 'explain' transcription and that a variety of transcription factors might have broad roles in general transcription levels across many genes

Modeling gene expression using chromatin features in various cellular context

Dong et al. (2012) *Genome Biology*. 12:R53

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Many variants in ENCODE-regions

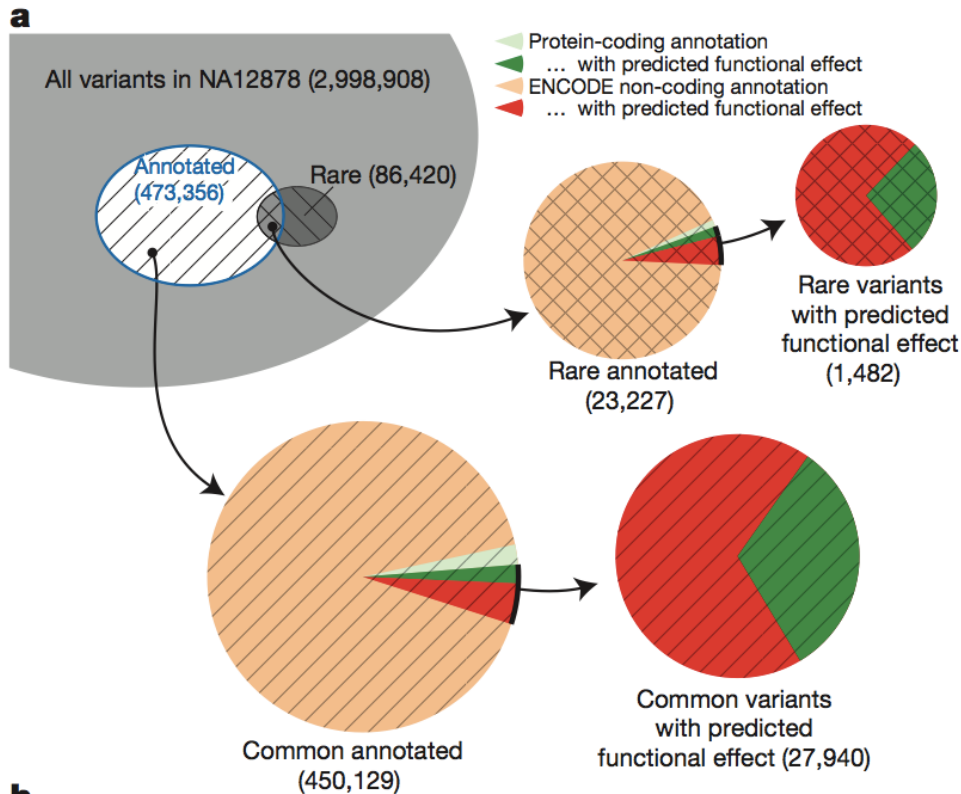


Figure 9 | Examining ENCODE elements on a per individual basis in the normal and cancer genome. **a**, Breakdown of variants in a single genome (NA12878) by both frequency (common or rare (that is, variants not present in the low-coverage sequencing of 179 individuals in the pilot 1 European panel of the 1000 Genomes project⁵⁵)) and by ENCODE annotation, including protein-coding gene and non-coding elements (GENCODE annotations for protein-coding genes, pseudogenes and other ncRNAs, as well as transcription-factor-binding sites from ChIP-seq data sets, excluding broad annotations such as histone modifications, segmentations and RNA-seq). Annotation status is further subdivided by predicted functional effect, being non-synonymous and missense mutations for protein-coding regions and variants overlapping bound transcription factor motifs for non-coding element annotations. A substantial proportion of variants are annotated as having predicted functional effects in the non-coding category. **b**, One of several relatively rare occurrences, where

Breakdown of variants by frequency

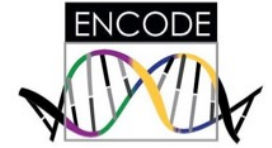
- Common or Rare (that is, variants not present in the low-coverage sequencing of 179 individuals in the pilot 1 European panel of the 1000 Genomes project)
- ENCODE annotation, including protein-coding gene and non-coding elements

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ENCODE and Disease

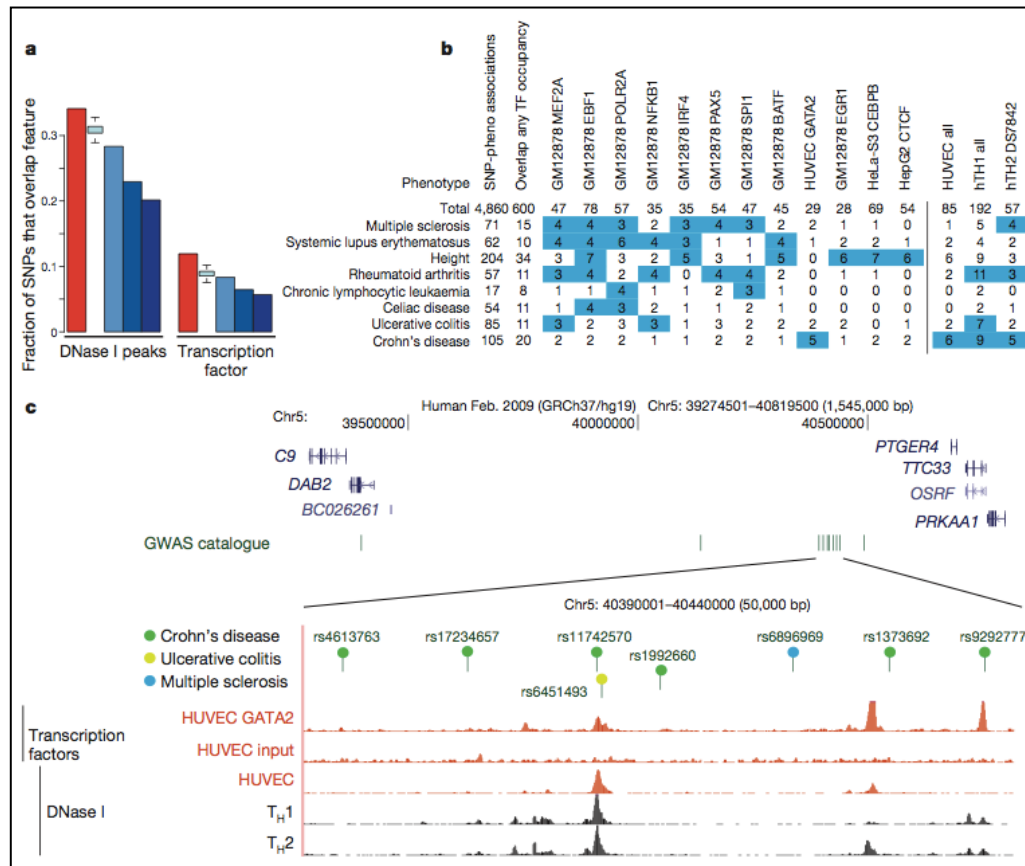
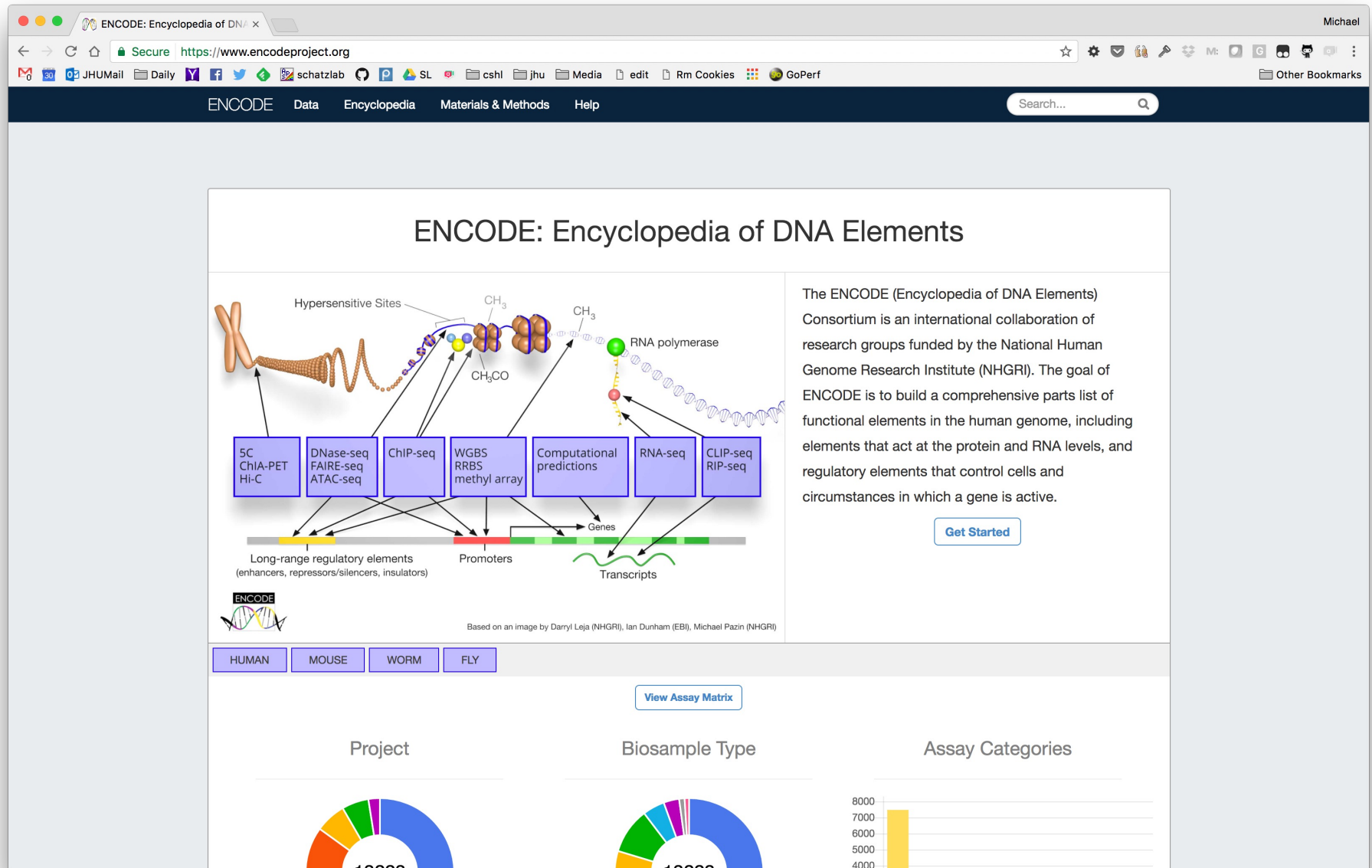


Figure 10 | Comparison of genome-wide-association-study-identified loci with ENCODE data. **a**, Overlap of lead SNPs in the NHGRI GWAS SNP catalogue (June 2011) with DHSs (left) or transcription-factor-binding sites (right) as red bars compared with various control SNP sets in blue. The control SNP sets are (from left to right): SNPs on the Illumina 2.5M chip as an example of a widely used GWAS SNP typing panel; SNPs from the 1000 Genomes project; SNPs extracted from 24 personal genomes (see personal genome variants track at <http://main.genome-browser.bx.psu.edu> (ref. 80)), all shown as blue bars. In addition, a further control used 1,000 randomizations from the genotyping SNP panel, matching the SNPs with each NHGRI catalogue SNP for allele frequency and distance to the nearest TSS (light blue bars with bounds at 1.5 times the interquartile range). For both DHSs and transcription-factor-binding regions, a larger proportion of overlaps with GWAS-implicated SNPs is found compared to any of the controls sets. **b**, Aggregate overlap of

phenotypes to selected transcription-factor-binding sites (left matrix) or DHSs in selected cell lines (right matrix), with a count of overlaps between the phenotype and the cell line/factor. Values in blue squares pass an empirical P -value threshold ≤ 0.01 (based on the same analysis of overlaps between randomly chosen, GWAS-matched SNPs and these epigenetic features) and have at least a count of three overlaps. The P value for the total number of phenotype-transcription factor associations is < 0.001 . **c**, Several SNPs associated with Crohn's disease and other inflammatory diseases that reside in a large gene desert on chromosome 5, along with some epigenetic features indicative of function. The SNP (rs11742570) strongly associated to Crohn's disease overlaps a GATA2 transcription-factor-binding signal determined in HUVECs. This region is also DNase I hypersensitive in HUVECs and T-helper T_H1 and T_H2 cells. An interactive version of this figure is available in the online version of the paper.

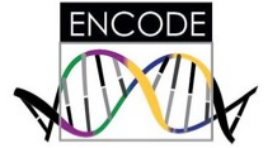
- 88% of GWAS SNPs are intronic or intergenic of unknown function
- We found that 12% of these GWAS-SNPs overlap transcription-factor-occupied regions whereas 34% overlap DHSs
- GWAS SNPs are particularly enriched in the segmentation classes associated with enhancers and TSSs across several cell types

ENCODE Studies



>5000 Citations for main paper; >>10k for all papers

Summary & Critique



- **Summary**

- *The unprecedented number of functional elements identified in this study provides a valuable resource to the scientific community as well as significantly enhances our understanding of the human genome.*

- **Critique**

- Was it correct?
- What is functional?
- What is conservation?
- What was the control?
- What are the tradeoffs of organizing so much funding (\$288M!) around a single project; will other groups successfully use these data?

Redefining the Nature of the Gene



1865 Gregor Mendel defines quantitative traits

1911 Thomas Hunt Morgan
links genes to chromosomes

1943 Max Delbrück and Salvador Luria
show mutations can pre-exist in genes

1953 James D. Watson, Francis Crick, Maurice Wilkins, and
Rosalind Franklin determine the structure of DNA, pointing
implicitly to the mechanism of genetic inheritance

1961 Francois Jacob and Jacques Monod define the
nature of genes as regulated linear elements
on the chromosomes of bacteria

1963 Marshall Warren Nirenberg, Har Gobind Khorana
and Robert William Holley break the genetic code:
they can now read a DNA sequence and use it
to predict a protein's amino acid sequence

1977 Phillip Sharp and Richard Roberts discover
"split genes": the fact that coding portions of most
genes are interrupted by non-coding portions

2001 First draft of human genome is generated
by the Human Genome Project, revealing
only 1%-2% encodes proteins

2002-2012
ENCODE project shows most of the
genome is transcribed, prompting
a new definition for the gene

Comment on “Evidence of Abundant Purifying Selection in Humans for Recently Acquired Regulatory Functions”



Phil Green* and Brent Ewing

Ward and Kellis (Reports, 28 September 2012, p. 1675; published online 5 September 2012) found altered patterns of human polymorphism in biochemically active but non-mammalian-conserved genomic regions relative to control regions and interpreted this as due to lineage-specific purifying selection. We find on closer inspection of their data that the polymorphism trends are primarily attributable to mutational variation and technical artifacts rather than selection.

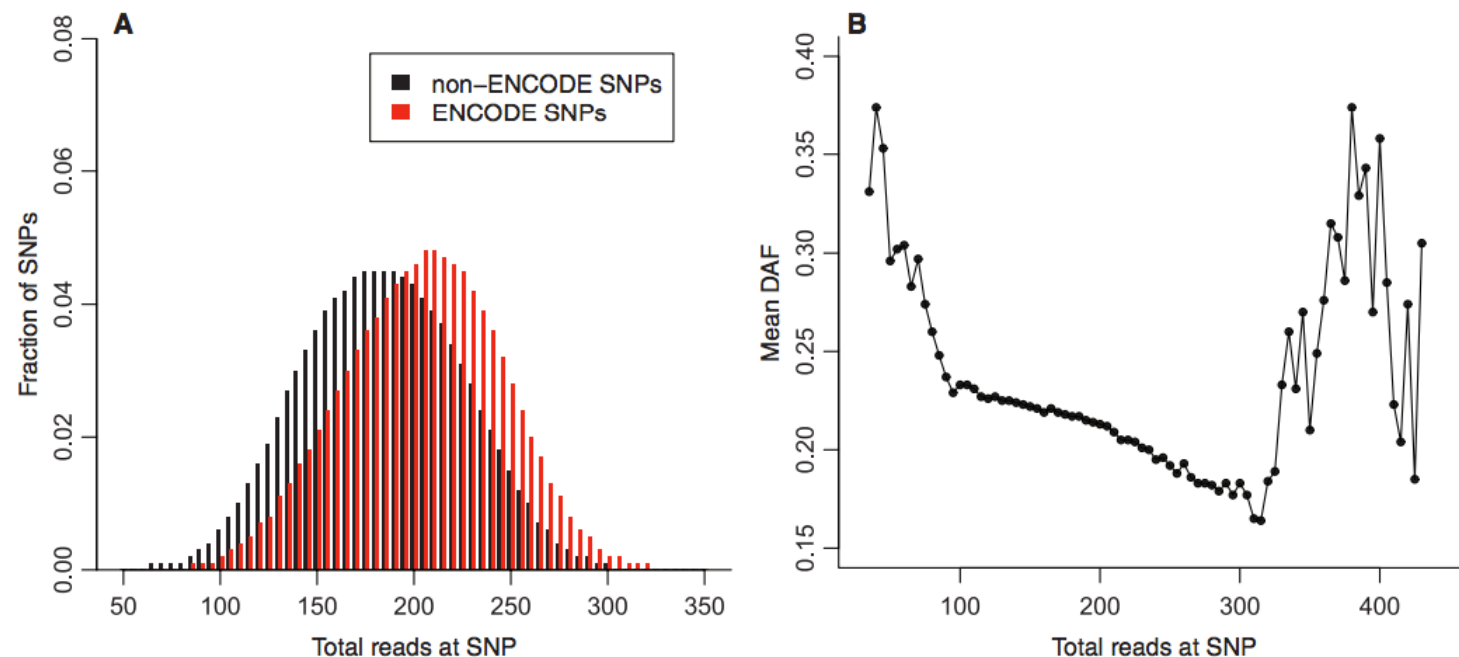


Fig. 1. Variation in 1000 Genomes read depth (totaled over 59 Yoruban individuals) and its impact on DAF. (A) Read-depth distribution for SNPs in neutral control (non-ENCODE) and ENCODE target regions. **(B)** DAF as a function

of read depth, for non-ENCODE SNPs. DAF decreases with increasing depth, due to increasing sensitivity to detect rare variants; the reverse trend at depths above 300 likely reflects the presence of spurious “paralogue-collapse” SNPs.

On the Immortality of Television Sets: “Function” in the Human Genome According to the Evolution-Free Gospel of ENCODE

Dan Graur^{1,*}, Yichen Zheng¹, Nicholas Price¹, Ricardo B.R. Azevedo¹, Rebecca A. Zufall¹, and Eran Elhaik²

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Accepted: February 16, 2013

Abstract

A recent slew of ENCyclopedia Of DNA Elements (ENCODE) Consortium publications, specifically the article signed by all Consortium members, put forward the idea that more than 80% of the human genome is functional. This claim flies in the face of current estimates according to which the fraction of the genome that is evolutionarily conserved through purifying selection is less than 10%. Thus, according to the ENCODE Consortium, a biological function can be maintained indefinitely without selection, which implies that at least $80 - 10 = 70\%$ of the genome is perfectly invulnerable to deleterious mutations, either because no mutation can ever occur in these “functional” regions or because no mutation in these regions can ever be deleterious. This absurd conclusion was reached through various means, chiefly by employing the seldom used “causal role” definition of biological function and then applying it inconsistently to different biochemical properties, by committing a logical fallacy known as “affirming the consequent,” by failing to appreciate the crucial difference between “junk DNA” and “garbage DNA,” by using analytical methods that yield biased errors and inflate estimates of functionality, by favoring statistical sensitivity over specificity, and by emphasizing statistical significance rather than the magnitude of the effect. Here, we detail the many logical and methodological transgressions involved in assigning functionality to almost every nucleotide in the human genome. The ENCODE results were predicted by one of its authors to necessitate the rewriting of textbooks. We agree, many textbooks dealing with marketing, mass-media hype, and public relations may well have to be rewritten.

Key words: junk DNA, genome functionality, selection, ENCODE project.

The ENCODE project: Missteps overshadowing a success

Two clichés of science journalism have now played out around the ENCODE project. ENCODE's publicity first presented a misleading "all the textbooks

*"To clarify what noise means, I propose the **Random Genome Project**. Suppose we put a few million bases of entirely random synthetic DNA into a human cell, and do an ENCODE project on it. Will it be reproducibly transcribed into mRNA-like transcripts, reproducibly bound by DNA-binding proteins, and reproducibly wrapped around histones marked by specific chromatin modifications? I think yes.*

A striking feature of genetic regulation is that regulatory factors (proteins or RNAs) generally recognize and bind to small sites, small enough that any given factor will find specific binding sites even in random DNA. Promoters, enhancers, splice sites, poly-A addition sites, and other functional features in the genome all have substantial random occurrence frequencies. These sites are not nonspecific in a random genome. They are specific sequences, albeit randomly occurring and not under selection for any function.

Would biochemical activities in the random genome be regulated under different conditions? For example, would they be cell type-specific? Surely yes, because the regulatory factors themselves (such as transcription factors) are regulated and expressed in specific cell types and conditions."



Bruce Alberts is Editor-in-Chief of *Science*.

The End of “Small Science”?

I AM PROMPTED TO WRITE THIS EDITORIAL BY THE RELEASE OF 30 PAPERS THIS MONTH FROM THE ENCODE Project Consortium. This decade-long project involved an international team of 442 scientists who have compiled what is being called an “encyclopedia of DNA elements,” a comprehensive list of functional elements in the human genome. The detailed overview is expected to spur further research on the fundamentals of life, health, and disease. ENCODE exemplifies a “big-science” style of research that continues to sweep the headlines, and the increased efficiency of data production by such projects is impressive. Does this mean that the highly successful “small-science” era of biological research will soon be over? Will government funding increasingly favor big-science projects? I certainly hope that the answer is no.

...

Each year, the amount of factual information that scientists acquire about cells increases and, stimulated by -omics projects, the compilations of data expand at a tremendous rate. But the grand challenges that remain in attaining a deep understanding of the chemistry of life will require going beyond detailed catalogs. Ensuring a successful future for the biological sciences will require restraint in the growth of large centers and -omics-like projects, so as to provide more financial support for the critical work of innovative small laboratories striving to understand the wonderful complexity of living systems.

— Bruce Alberts

10.1126/science.1230529